

Preliminary Design Review: FFVTOL (VTOC)

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This Preliminary Design Review (PDR) report evaluates a solar-augmented, fixed-wing eVTOL UAS derived from the Stallion platform. The design introduces three upgrades: flexible photovoltaic wing film targeting $\geq 20\%$ endurance gain; a quick-swap modular nose for EO/IR/thermal/night-vision payloads; and GPS-independent navigation using a CSAC-disciplined INS with visual/RF localization. The review defines mission architecture for reconnaissance, disaster response, and public safety; establishes requirements, interfaces, and risk mitigations; and reports aerodynamic, power-balance, and navigation-in-the-loop analyses. Results indicate L/D preserved within 5% of baseline, $\leq 3\%$ drift in GPS-denied operation, < 5 -min payload swaps, and FAA Part 107 compliance. Collectively, the configuration and verification plan demonstrate readiness for Critical Design Review.

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Nomenclature

A = Effective solar area or rotor disk area

AR = Aspect Ratio

AoA = Angle of Attack

A_{rotor} = Rotor disk area

A_{solar} = Effective solar area

$CASEVAC$ = Casualty Evacuation

CCD = Charge-Coupled Device

CD = Drag coefficient

CD_{front} = Frontal drag coefficient

$CD_{induced}$ = Induced drag coefficient

CL = Lift Coefficient

CL_{Max} = Maximum Lift Coefficient

CG = Center of Gravity

D = Drag force

da/dN = Crack growth rate

DoD = Depth of Discharge

d_i = Distance in trilateration

ΔK = Stress intensity factor range

ΔT = Temperature difference

E = Energy

EKF = Extended Kalman Filter

e = Oswald efficiency factor or error

FF = Fill factor

FM = Figure of Merit

FOV = Field of View

F_{ext} = External forces

g = Gravitational acceleration

GSD = Ground Sampling Distance

GTOW = Gross Takeoff Weight

H = Measurement matrix

h = Heat transfer coefficient

I = Solar irradiance or current or impact

I₀ = Saturation current

I_{ph} = Photocurrent

I_{sc} = Short-circuit current

IMU = Inertial Measurement Unit

k = Boltzmann constant

K = Kalman gain

KV = Constant Velocity

L = Lift Force

L/D = Lift-to-Drag ratio

LPR = License Plate Recognition

LWIR = Long-Wave Infrared

MAC = Mean Aerodynamic Chord

M_{ext} = External moments

m = Vehicle mass

n = Number of rotors or ideality factor

OIS = Optical Image Stabilization

P = Power or probability

P_{avg} = Average power

P_{hover} = Hover power

P_i = Induced power

PDR = Packet Delivery Ratio

P_s = Solar power generation

PWM = Pulse Width Modulation

Q = Heat flux or process noise

R = Radiance

Re = Reynolds number

RMS = Root Mean Square

S = Wing Reference area

$S-N$ = Stress-Number curve

SNR = Signal-to-Noise Ratio

SOC = State-of-Charge

T = Thrust or temperature

T/W = Thrust-to-Weight ratio

$TDOA$ = Time-Difference-of-Arrival

T_{per} = Thrust per rotor

$URLLC$ = Ultra-Reliable Low-Latency Communication

V = Airspeed or voltage

V_{oc} = Open-circuit voltage

V_{stall} = Stall speed

$VNIR$ = Visible Near-Infrared

W = Vehicle weight

$x_{err}, y_{err}, z_{err}$ = Position errors

ζ = Damping ratio

η = Solar cell efficiency or efficiency

η_d = Discharge efficiency

η_{prop} = Propeller efficiency

θ = Pitch angle or incidence angle

λ = Eigenvalue

μ = Viscosity

ν = Kinematic viscosity

ρ = Air Density

σ = Stress or Stefan-Boltzmann constant

ω = Angular velocity

Δt_i = Time difference

$\int P dt$ = Integrated power

I. Introduction

A. Executive Summary

The purpose of this Preliminary Design Review (PDR) report is to present the preliminary design results for the design of a UAV with flexible photovoltaic wing film; a quick-swap modular nose for EO/IR/thermal/night-vision payloads; and GPS-independent navigation using a CSAC-disciplined INS with visual/Rf localization, developed by Team 5 in fulfillment of the requirements of AEM 402: Integrated Aerospace Design (Senior Design) at The University of Alabama. This document demonstrates that the preliminary design meets system requirements with acceptable risk, within cost and schedule constraints, and establishes the technical foundation for proceeding to the critical design phase.

Our team, composed of Fourie van Rooyen, Eliel Smith, Matthew Cotting, Cohen McNabb, Coralie Lallier, and Colin Thomasma, has worked collectively under the guidance of Dr. Jinwei Shen to define mission objectives, generate and evaluate alternative concepts, down-select the optimal design, and establish the baseline architecture, operations concept, and subsystem specifications

B. Governing Equations

The governing equations for the FFVTOL UAV design are consolidated from individual contributions across aerodynamics, propulsion, power systems, stability, control, and transition dynamics. These equations underpin the feasibility analyses for reconnaissance missions, ensuring the vehicle meets requirements for VTOL capability, cruise efficiency, stealth, and endurance in diverse environments such as military intelligence, firefighting, and law enforcement surveillance. They are presented in a flowing narrative for clarity, with derivations, assumptions, sample calculations, and relevance to the mission provided where applicable.

Core aerodynamic performance relies on established equations. For cruise lift, the equation is:

$$L = \frac{1}{2} \rho V^2 S C_L \quad (1)$$

Air density is 1.225 kilograms per cubic meter at sea level, cruise speed is 20 meters per second, wing area is 0.5 square meters, and nominal lift coefficient is 1.0 for a NACA 4412 airfoil. Derived from Bernoulli's theorem and momentum flux across a control volume, this calculates the lift force. Sample calculation: dynamic pressure equals 245 pascals, lift equals 122.5 newtons, exceeding weight of 98.1 newtons by 25 percent for maneuver margins. Analysis of this relation ensures sufficient lift during fixed-wing cruise for extended reconnaissance, providing stability in loitering phases over targets like fire perimeters or urban surveillance areas.

Drag estimation complements lift analysis. For total drag, the equation is:

$$D = \frac{1}{2} \rho V^2 S C_D \quad (2)$$

Parasitic drag coefficient is 0.02, Oswald efficiency is 0.8, aspect ratio is 4.5. Derived parasitically from skin friction and induced from lifting line theory. Sample: induced coefficient equals 0.088, total coefficient approximately 0.108, drag approximately 13.23 newtons. Analyzing drag to size ratio allows for efficient propulsion for cruise, minimizing power draw during long-endurance missions in GPS-denied or contested environments.

To verify flow regime, the dimensionless parameter is:

$$R_e = \frac{\rho V c}{\mu} \quad (3)$$

This checks operation in the low-Reynolds regime, informing boundary-layer behavior and CFD turbulence models. Guiding airfoil selection ensures aerodynamic predictions align with real-world flow conditions for reliable performance in variable atmospheres, such as smoke-filled firefighting scenarios.

For efficient cruise lift coefficient, the relation is:

$$C_L = \frac{W}{qS}, \quad q = \frac{1}{2} \rho V^2 \quad (4)$$

This ensures the wing generates sufficient lift in steady cruise by adjusting area and velocity to keep the coefficient efficient, optimizing the planform for balanced flight, supporting modular payload weights in reconnaissance tasks without compromising endurance.

Complementing drag breakdown, the full drag coefficient includes induced effects:

$$C_D = C_{D0} + \frac{C_L^2}{\pi e A R}, \quad D = q S C_D, \quad P = \frac{DV}{\eta_p} \quad (5)$$

This establishes drag and power needs in forward flight as it is essential for sizing the cruise propulsor and checking whether endurance goals are achievable within the system's energy budget.

For ideal induced velocity in hover, the momentum theory gives:

$$v_i = \sqrt{\frac{T_i}{2\rho A_i}}; \quad P_{i(ideal)} = T_i v_i = \frac{T_i^{\frac{3}{2}}}{2\rho A_i} \quad (6)$$

This governs hover performance; summing across all rotors and applying corrections yields installed hover power and ceiling margins.

Building on induced velocity, the ideal power for hover per rotor is:

$$P_i = \frac{T_i^{\frac{3}{2}}}{\sqrt{2\rho A}} \quad (7)$$

Total thrust is 98.1 newtons, 4 rotors, disk area approximately 0.0707 square meters. Sample: power approximately 292 watts per rotor, total 1168 watts, scaled by 0.7 to about 1670 watts. This equation helps to determine power needs for VTOL in confined spaces, enabling launches from restricted areas like urban clearings or fire lines, with redundancies for fault-tolerant operations.

For propeller thrust coefficient, the nondimensional form is:

$$C_T = \frac{T}{\rho n^2 D^4} \quad (8)$$

This enables scaling across geometries and guides rotor design for efficient thrust across regimes.

Similarly, for power coefficient:

$$C_p = \frac{P}{\rho n^3 D^5} \quad (9)$$

This complements thrust scaling and assists in optimizing diameter, pitch, and RPM for efficient operations.

Lastly, propeller efficiency is captured by:

$$\eta_p = \frac{TV}{P} \quad (10)$$

This relation ties thrust, speed, and power to ensure energy-efficient propulsion during reconnaissance phases.

For static longitudinal stability, the derivative must satisfy:

$$C_{m\alpha} < 0 \quad (11)$$

This ensures negative pitching moment with increasing angle of attack to maintain vehicle response to disturbances, supporting safe flight in turbulent conditions during firefighting or military missions.

Complementing stability, the static margin is:

$$SM = \frac{x_{np} - x_{cg}}{MAC} \times 100 [\%] \quad (12)$$

This quantifies stability margin; it confirms controllability across the center of gravity envelope during transitions, essential for autonomous operations in unpredictable environments like jammed GPS zones. Overall, these nondimensional parameters allow for consistent scaling of different propeller geometries, informing the selection of diameter, pitch, and RPM to maximize thrust efficiency, with results validated through MATLAB analyses and Ansys Fluent simulations.

For solar power generation during flight:

$$P_s = \eta I A \quad (13)$$

Efficiency is 0.2, irradiance is 1000 watts per square meter, area is 0.3 square meters, yielding about 60 watts. The battery is supplemented for around 10-20 percent endurance extension under solar conditions, critical for prolonged loitering in remote reconnaissance without recharging infrastructure.

To smooth transition commands, the scheduling function is:

$$\lambda(s) = 3s^2 - 2s^3 \in [0, 1] \quad (14)$$

This provides a cubic blend from 0 to 1, optimizing the transition path for minimal disturbance, vital for stealth and endurance in multi-phase missions.

Based on scheduling, the command for the angle of attack is:

$$\alpha_{cmd}(t) = \lambda(s)\alpha_{FW} \quad (15)$$

This interpolates to fixed-wing value which ensures gradual shift during mode changes for stable sensor data collection.

For attitude dynamics modeling:

$$\ddot{a} + 2\zeta_a \omega_a \dot{a} + \omega_a^2 a = w_a^2 a_{cmd}, \text{ where } a = \alpha \quad (16)$$

Models control response for stable hover and transition, tuned for damping to prevent oscillations in dynamic environments.

Control allocation differs by mode. For vertical take-off and landing inputs:

$$\text{VTOL: } T = u_T, \phi_c = u_\phi, \theta_c = -u_\theta, \psi_c = u_\psi \quad (17)$$

This maps to thrust and attitudes and defines actuator commands for vertical operations in confined spaces.

For fixed-wing:

$$\text{FW: } T = U_T, \text{ail} = u_{\text{ail}}, \text{elev} = -u_{\text{elev}}, \text{rud} = u_{\text{rud}} \quad (18)$$

This maps to surfaces, which enables precise maneuvering in cruise for effective reconnaissance.

Fundamental to VTOL capability is the thrust-to-weight ratio:

$$\frac{T}{W} \geq 1.0 \quad (19)$$

The optimal value for this ratio is 1.3-1.5. Such a value ensures rapid ascent and payload adaptability in emergency responses, like military casualty evacuation or firefighting, providing power margins for demanding environments.

Power for hovering follows momentum theory:

$$P_{\text{hover}} = \frac{W^{\frac{3}{2}}}{\sqrt{2\rho A}} \quad (20)$$

This determines demands, driving distributed propulsion with multiple rotors to reduce demands and add safety redundancies.

During transition, lift balance is:

$$L_{\text{total}} = L_{\text{rotor}} + L_{\text{wing}} = W \quad (21)$$

This ensures equilibrium. Said equilibrium enables seamless VTOL to cruise shifts, reducing workload in autonomous reconnaissance

C. Project Statement and Objectives

The mission of this project is to design and validate a fixed-wing, fully autonomous reconnaissance unmanned aerial vehicle (UAV) capable of long-endurance operations for live video footage, precision imaging, and threat assessment. The UAV must meet stakeholder-defined constraints of affordability, safety, manufacturability, and mission reliability.

The primary objectives are to:

1. Develop a UAV capable of autonomous flight for extended endurance (target: 60 Minutes) with real-time crop monitoring capability.
2. Ensure compliance with FAA Part 107 operational requirements.
3. Design for modularity, manufacturability, and cost-effectiveness.

Secondary objectives include demonstrating technical innovation in avionics integration, robustness against environmental disturbances, and compliance with NASA NPR 7123.1D systems engineering methodology.

D. Sponsoring Agencies

This project is supported by The University of Alabama Department of Aerospace Engineering and Mechanics.

Fly Sheet (Key Characteristics)

Category	Preliminary Value / Description
Mission Concept	Fixed-wing autonomous UAV for reconnaissance missions
Configuration	High-aspect ratio fixed-wing aircraft with electric propulsion
Wingspan	1.33 m
Maximum Takeoff Weight (MTOW)	5 kg
Endurance	~ 1 hour
Cruise Speed	17 m/s (37-43 mph)
Payload	Multispectral imaging system (≤ 2 kg)
Mission Environment	Low-altitude flight (≤ 400 ft AGL), slight winds/rain
Subsystems	Airframe, propulsion, avionics, GNC, communications, payload
Design Status	Preliminary design completed; trade studies and down-select finalized
Schedule Status	On track with course milestones; critical design scheduled for Spring 2026
Risk Posture	Acceptable with identified mitigations in propulsion endurance, avionics integration, and payload imaging performance

E. Summary

This report demonstrates that the preliminary design has matured sufficiently to justify transition into the critical design phase. The chosen concept satisfies the mission requirements, adheres to stakeholder constraints, and has been evaluated through trade studies and preliminary performance analyses. Critical risks have been identified, and mitigation plans are in place. The design possesses adequate technical margins in mass, power, and structural capability to proceed.

The team concludes that the project is feasible, the preliminary design is robust, and that sufficient resources and plans exist to continue development into the detailed design stage.

II. Purpose and Scope (SEMP)

A. Purpose

The purpose of this project is to design and validate a fixed-wing, fully autonomous unmanned aerial vehicle (UAV) capable of performing long-endurance reconnaissance missions to support intelligence, surveillance, and operational awareness. Modern security and defense environments are increasingly data-driven, requiring timely and accurate information regarding terrain, personnel movement, and environmental conditions. Remote sensing with high-resolution imaging and specialized sensor payloads provides an effective solution, yet the cost and complexity of existing reconnaissance platforms remain a barrier to widespread deployment.

This project seeks to address these challenges by producing a UAV platform that is affordable, manufacturable with readily available materials, compliant with regulatory requirements, and capable of sustained autonomous operation. By integrating advanced avionics, efficient propulsion, and a modular payload architecture, the system will enhance mission planners' and operators' ability to gather actionable intelligence, improve situational awareness, and support informed decision-making during reconnaissance operations.

The PDR represents a major milestone in the systems engineering life cycle, demonstrating that the preliminary design meets all system-level requirements, identifies and mitigates risks, and establishes a credible foundation for the Critical Design Review (CDR). Specifically, this PDR ensures that:

- The mission objectives are well-defined and achievable.
- A feasible system architecture has been developed and down-selected through trade studies.
- Subsystems have been baselined with sufficient detail to confirm technical viability.
- Risks have been identified and mitigation strategies outlined.
- Adequate programmatic resources (schedule, cost, and facilities) are available.

This purpose is consistent with the NASA NPR 7123.1D systems engineering framework, ensuring that design maturity aligns with expectations for a PDR-level review.

C. Scope

The scope of this project encompasses the formulation and preliminary design phases of the UAV. Work completed to date includes mission definition, requirements analysis, concept of operations development, trade study evaluations, subsystem sizing, and integration planning.

The system under development is a fixed-wing UAV with the following baseline parameters, derived from MCR trade studies:

- **Wingspan:** 1.33 m
- **Maximum Takeoff Weight (MTOW):** 7 kg
- **Cruise Speed:** 17 m/s
- **Endurance:** ~ 1 hours
- **Payload Capacity:** ≤ 2.0 kg (multispectral imaging system)
- **Operating Altitude:** ≤ 400 ft AGL (in compliance with FAA Part 107)

The scope of this design does **not** extend to full-scale flight testing or operational deployment. Instead, it is limited to the academic demonstration of feasibility through system-level design, subsystem analyses, and verification/validation planning. The PDR will define the configuration baseline to be carried forward into CDR, where subsystem designs will be finalized and detailed engineering drawings will be completed.

Key boundaries of the project include:

- **Regulatory Scope:** Limited to FAA Part 107 operations, excluding higher-altitude or beyond-visual-line-of-sight (BVLOS) operations.
- **Mission Scope:** Encompasses all potential applications, but mainly for reconnaissance missions, urban operations, search and rescue, police operations, and firefighting.
- **Technical Scope:** Focused on airframe, propulsion, avionics, and payload integration; manufacturing processes are evaluated at a conceptual level, not industrial scale.
- **Programmatic Scope:** Limited by course schedule and academic resources, with the project expected to reach CDR readiness by Spring 2026.

D. Systems Engineering Alignment

This PDR report follows the Systems Engineering Management Plan (SEMP) structure outlined in NPR 7123.1D, ensuring alignment with NASA systems engineering best practices. The purpose of this alignment is to demonstrate that the team is applying structured design processes, including requirements traceability, risk management, verification and validation planning, and lifecycle reviews.

The scope of the report includes:

- Defining mission objectives and program success criteria.
- Establishing baseline system architecture and concept of operations.
- Performing trade studies to justify design decisions.
- Developing subsystem specifications and integration plans.
- Outlining programmatic resources, schedule, and risk management strategies.

By adhering to these principles, the team ensures that the project not only satisfies academic requirements but also reflects industry-standard engineering practice.

E. Summary

In summary, the purpose of this project is to design a long-endurance UAV that supports a wide range of reconnaissance payloads for diverse applications, including but not limited to military operations, urban operations, search and rescue, police operations, and firefighting operations. The scope defines the mission boundaries, technical parameters, and programmatic constraints under which the design is developed. This Preliminary Design Review (PDR) confirms that the team has sufficiently defined the problem, established a feasible solution, and prepared the groundwork for detailed design, ensuring the UAV's adaptability and long-endurance capabilities meet the needs of multiple operational domains.

III. Mission Objectives (SEMP)

A. Overview

Mission objectives define the fundamental purpose of the project and serve as the basis for requirements development, system architecture, and success criteria. For this Preliminary Design Review (PDR), the objectives of the Fully Autonomous reconnaissance UAV for intelligence missions are divided into primary and secondary objectives. These objectives were established during the Mission Concept Review (MCR) and refined during the preliminary design phase through trade studies, stakeholder expectations, and program constraints.

The objectives are designed to be:

- **Clear and measurable**, providing a basis for success criteria and verification.
- **Technically feasible**, achievable within the constraints of available resources and schedule.
- **Programmatically relevant**, aligned with the academic scope of AEM 402 and compliant with applicable regulatory frameworks.

B. Primary Objectives

The primary objectives represent the essential functions that must be satisfied for the mission to be successful. These objectives are directly linked to the system's high-level requirements and will guide the critical design phase.

1. Develop a long-endurance autonomous UAV platform

- The UAV must demonstrate a sustained endurance of ~1 hour at a nominal cruise speed of 17 m/s, enabling coverage of large areas in a single mission.
- This endurance objective is tied to the propulsion system efficiency, structural weight reduction, and aerodynamic optimization.

2. Integrate a precision multispectral imaging payload

- The system must be capable of carrying and operating a variety of interchangeable reconnaissance payloads, including but not limited to multispectral, thermal, LiDAR, and high-resolution imaging cameras, each ≤ 2.0 kg, to support diverse applications such as live video feed from missions (e.g., number of combatants, troop positioning and safety), military surveillance, search and rescue, police operations, firefighting, commercial cargo transport monitoring, urban operations, and other relevant use cases.
- Payload integration must ensure minimal impact on aerodynamic stability and endurance.

3. Ensure compliance with FAA Part 107 operational requirements

- The UAV must operate below 400 ft AGL, within line-of-sight restrictions, and remain within the weight and safety constraints defined by FAA regulations.
- This objective ensures the project remains realistic and legally deployable across a wide range of real-world operational settings, including military surveillance, search and rescue, police operations, firefighting, urban operations, and other relevant use cases, while accommodating diverse reconnaissance payloads.

4. Design for manufacturability and modularity

- The aircraft must be manufacturable with readily available materials and fabrication methods accessible within an academic environment.
- Subsystems should be modular to allow for rapid integration, testing, and potential future upgrades.

C. Secondary Objectives

The secondary objectives support broader project goals by enhancing reliability, maintainability, and technical credibility. These objectives are not critical for minimum mission success but contribute to project quality and stakeholder value.

1. Demonstrate advanced autonomy and robust avionics integration

- The UAV should implement waypoint-based navigation and autonomous flight modes.
- The avionics system must ensure robust communication, flight control stability, and safe return-to-home functionality in case of failure.

2. Optimize aerodynamic efficiency and structural performance

- The airframe should be designed to minimize drag and maximize lift-to-drag ratio, improving endurance margins.
- Structural analysis should ensure adequate safety factors against expected aerodynamic and ground loads.

3. Ensure environmental robustness

- The UAV must be capable of operating under variable environmental conditions, including wind gusts, moderate turbulence, and temperature variations, to ensure reliable performance across diverse applications

4. Follow NASA NPR 7123.1D systems engineering practices

- Requirements traceability, risk management, and verification/validation planning must be fully implemented to align with professional engineering standards.

D. Alignment with Stakeholder Expectations

Stakeholder expectations were defined during the MCR and refined in consultation with faculty guidance. These include:

- **Students (Team 5):** Gain hands-on experience in systems engineering, aircraft design, and project management.

- **Faculty (Sponsor/Reviewers):** Demonstrate technical feasibility, adherence to NASA systems engineering standards, and credible academic outcomes.
- **End Users (Defense Sector):** Provide a UAV concept that reduces operational costs, enhances reconnaissance capabilities, and is compliant with FAA regulations.

By fulfilling the mission objectives, the project will satisfy stakeholder needs while ensuring academic and technical rigor.

E. Summary

The mission objectives define the performance, safety, and programmatic boundaries of the UAV project. The primary objectives ensure the design satisfies mission-essential requirements, while the secondary objectives enhance system robustness and align with professional standards. These objectives will serve as the foundation for requirements allocation, subsystem design, and success criteria throughout the remainder of the project lifecycle.

IV. Program Success Criteria (SEMP)

A. Overview

Program success criteria define the quantitative and qualitative measures by which the project will be judged. They serve as the reference against which system performance is evaluated during the Preliminary Design Review (PDR), and later in the Critical Design Review (CDR). These criteria are directly derived from the mission objectives (Section 2) and aligned with stakeholder expectations, FAA Part 107 regulations, and NASA NPR 7123.1D systems engineering guidelines.

The success criteria are categorized into technical, programmatic, and operational domains. Each criterion has an associated measure of effectiveness (MOE), measure of performance (MOP), or technical performance measure (TPM) that defines how compliance will be evaluated.

B. Technical Success Criteria

1. Endurance Requirement

- *Success Threshold:* UAV demonstrates an endurance of at least 1 hour at a cruise speed of 17 m/s.
- *Evaluation Method:* Verified through performance calculations, validated by subsystem-level energy analysis, and ultimately confirmed via prototype flight testing.

2. Payload Integration Requirement

- *Success Threshold:* The UAV must support a variety of interchangeable reconnaissance payloads, including but not limited to multispectral, thermal, LiDAR, and high-resolution imaging cameras, each ≤ 2.0 kg, without exceeding the maximum takeoff weight (10.5 kg), ensuring compatibility with diverse applications
- *Evaluation Method:* Verified through mass budgeting and structural analysis; validated through integration mock-up and system-level testing.

3. Aerodynamic and Structural Performance Requirement

- *Success Threshold:* Airframe achieves a lift-to-drag ratio (L/D) sufficient to sustain cruise with required endurance margin; structural components demonstrate a minimum factor of safety ≥ 1.5 under expected load conditions.

- *Evaluation Method:* Verified through CFD analysis, structural FEM simulations, and analytical sizing methods.

4. **Autonomy and Navigation Requirement**

- *Success Threshold:* Avionics system executes waypoint-based navigation, maintains stability under moderate disturbances, and demonstrates autonomous return-to-home capability in case of communication loss.
- *Evaluation Method:* Verified through hardware-in-the-loop simulations; validated by prototype flight trials.

C. **Programmatic Success Criteria**

1. **Schedule Compliance**

- *Success Threshold:* Project milestones (MCR, PDR, CDR, final demonstration) are completed on schedule as outlined in the project Gantt chart.
- *Evaluation Method:* Tracked via project management tools and reviewed at each major gate.

2. **Cost and Resource Compliance**

- *Success Threshold:* Project remains within allocated budget (materials, avionics, payloads) and makes effective use of available academic facilities.
- *Evaluation Method:* Monitored via budget tracking sheets and expenditure logs.

3. **Systems Engineering Adherence**

- *Success Threshold:* Requirements are traceable from mission objectives down to subsystem specifications, with TBD/TBR items resolved before CDR.
- *Evaluation Method:* Evaluated through requirements documentation, traceability matrices, and review board assessment.

D. **Operational Success Criteria**

1. **Regulatory Compliance**

- *Success Threshold:* UAV operates within FAA Part 107 constraints: ≤ 400 ft AGL, ≤ 55 lbs MTOW, line-of-sight operations, and safe recovery procedures.
- *Evaluation Method:* Confirmed through design analysis and compliance documentation.

2. **Safety and Risk Management**

- *Success Threshold:* All identified high-priority risks (propulsion endurance, avionics integration, payload data reliability) have defined mitigation strategies; residual risk is acceptable.
- *Evaluation Method:* Verified by risk matrix evaluation and reviewed during design reviews.

3. **Stakeholder Satisfaction**

- *Success Threshold:* The final design satisfies stakeholder expectations: affordable, reliable, and capable UAV concept for reconnaissance applications.
- *Evaluation Method:* Assessed qualitatively through faculty review and quantitatively through meeting defined MOEs and MOPs.

E. Summary

The program success criteria ensure that the UAV can be evaluated objectively against its mission objectives. These criteria provide the foundation for verification and validation activities, guiding subsystem refinement in preparation for CDR. By meeting the endurance, payload, regulatory, safety, and systems engineering requirements outlined here, the project will demonstrate technical feasibility, academic rigor, and readiness for progression into detailed design.

V. Applicable Documents (All)

A. Overview

Applicable documents establish the authoritative references that govern the design, evaluation, and management of the UAV. These documents include regulatory requirements, standards, and academic guidelines that define constraints, success criteria, and deliverable expectations. They serve as the baseline against which the design must demonstrate compliance and provide the framework for verification and validation activities.

B. Governing Systems Engineering Standards

1. NASA NPR 7123.1D – Systems Engineering Processes and Requirements

- Provides the overarching framework for lifecycle reviews, requirements traceability, technical planning, and verification/validation.
- Defines the Preliminary Design Review (PDR) entrance and success criteria used as the benchmark for this report.
- Ensures that the project follows structured systems engineering practices consistent with professional aerospace engineering standards.

2. NASA/SP-2016-6105 Rev2 – Systems Engineering Handbook

- Offers practical guidance for applying systems engineering principles.
- Informs the team's approach to requirements decomposition, risk management, and concept of operations development.

C. Regulatory and Operational Standards

1. FAA Part 107 – Small Unmanned Aircraft Regulations

- Governs the operational environment of UAVs ≤ 55 lbs in the United States.
- Defines restrictions relevant to this project, including:
 - Maximum altitude of 400 ft AGL.

- Line-of-sight operations.
- Operational weight limit ≤ 55 lbs.
- Requirements for safe recovery procedures.
- Compliance with Part 107 ensures that the design is realistic, legally deployable, and aligned with end-user expectations.

2. University of Alabama Institutional Requirements

- Project deliverables and design reviews are subject to AEM 402 course standards.
- Includes milestone deadlines (MCR, PDR, CDR), report formatting guidelines (AIAA style), and presentation requirements.

D. Technical and Design Standards

1. AIAA Technical Paper Format

- Establishes formatting requirements for technical reports, including margins, fonts, figure sizes, and citation style.
- Ensures professional presentation consistent with aerospace industry and research practices.

2. Avionics and Communication Standards

- Industry guidelines for UAV avionics integration (IEEE 802.15.4/Zigbee for telemetry, GPS protocols for navigation, and RC system redundancy).
- Compliance ensures safe and robust data communication between subsystems.

3. Structural and Materials Standards

- Reference to ASTM standards for small UAV composite and structural material testing.
- Guides preliminary structural sizing and ensures conservative safety factors.

E. Academic and Project-Specific Documents

1. Group Mission Concept Review (MCR) Report – September 2025

- Defined the mission need, objectives, and baseline concept.
- Serves as the foundation for PDR-level subsystem refinement and trade study justification.

2. AEM 402 Course PDR Memo (September 24, 2025) PDR Memo

- Defines the content, structure, and evaluation criteria for the PDR report.
- Provides required subsections, formatting, and page-length guidance.

3. Team Work Breakdown Structure (WBS) Documentation

- Provides the basis for programmatic planning, task allocation, and subsystem responsibilities.

F. Summary

The applicable documents form the regulatory, technical, and academic foundation of the UAV project. Compliance with FAA Part 107, NASA NPR 7123.1D, and AEM 402 course standards ensures that the preliminary design is not only technically feasible but also programmatically valid and legally operable. These references will continue to guide subsystem refinement, risk management, and verification planning as the project advances toward CDR.

VI. Technical Summary

A. Overview

The technical summary provides a consolidated description of the selected preliminary design for the Fully Autonomous Long-Endurance UAV. Following the Mission Concept Review (MCR), the team performed trade studies and system-level analyses to evaluate multiple candidate designs against mission objectives, stakeholder expectations, and program constraints. The outcome of this process was the selection of a fixed-wing, high-aspect-ratio UAV powered by an electric propulsion system. This configuration was determined to provide the best balance of endurance, payload capacity, manufacturability, and operational safety.

B. Trade Study Process

Three broad configurations were evaluated during the concept development stage:

1. Multirotor UAV

- Advantages: high maneuverability, vertical takeoff and landing (VTOL) capability.
- Disadvantages: limited endurance (<45 minutes), low aerodynamic efficiency, high power consumption.

2. Hybrid VTOL-Fixed Wing UAV

- Advantages: VTOL flexibility combined with improved cruise efficiency.
- Disadvantages: increased mechanical complexity, higher weight, reduced reliability, and increased cost.

3. Conventional Fixed-Wing UAV w tilt rotors (*Selected*)

- Advantages: high aerodynamic efficiency, long endurance (target: 1 hour), lower mechanical complexity, reduced weight.
- Disadvantages: tilt-rotor complexity, transition complexity, low maneuverability

A weighted trade study, considering endurance, manufacturability, payload integration, and regulatory compliance, indicated that the fixed-wing UAV was the superior solution.

C. Baseline Design Characteristics

The selected design baseline is summarized below (values drawn from MCR analysis and refined in PDR-level work):

- **Configuration:** High-aspect ratio, fixed-wing airframe with electric propulsion.
- **Wingspan:** 1.33 m.
- **Maximum Takeoff Weight (MTOW):** 7 kg.
- **Cruise Speed:** 17 m/s.
- **Endurance:** 1 hour.
- **Payload Capacity:** ≤ 2.0 kg imaging systems.
- **Operating Altitude:** ≤ 400 ft AGL (FAA Part 107 compliant).
- **Power System:** Electric motor with lithium-polymer battery pack sized for endurance mission profile.
- **Airframe Materials:** Lightweight composites and foam-core construction for manufacturability.

This baseline was selected for its ability to achieve endurance requirements while maintaining structural simplicity and compliance with operational constraints.

D. Subsystem Status

At the PDR stage, subsystem baselines have been established:

- **Structures:** Airframe design optimized for aerodynamic efficiency with structural safety factor ≥ 1.5 .
- **Propulsion:** Electric motor and battery sizing completed to achieve 1-hour endurance with margin.
- **Avionics & GNC:** Autopilot system selected; supports waypoint navigation, telemetry, and return-to-home capability.
- **Payload:** Multispectral camera identified and integrated into mass and power budgets.
- **Communications:** Telemetry system designed to maintain operator situational awareness within FAA Part 107 line-of-sight restrictions.

E. Risk and Margin Summary

The technical evaluation identified several key risks and associated mitigation strategies:

- **Endurance Risk:** Battery sizing and aerodynamic efficiency are critical to meeting endurance goals. *Mitigation:* maintain $\geq 10\%$ energy margin in sizing and validate performance with subsystem-level testing.
- **Avionics Integration Risk:** Potential instability in autonomous navigation under environmental disturbances. *Mitigation:* incorporate hardware-in-the-loop testing and redundant return-to-home functionality.
- **Payload Integration Risk:** Risk of adverse CG shift or drag penalties from payload placement. *Mitigation:* modular payload bay designed to minimize aerodynamic disruption.

Margins in mass, power, and structure have been maintained to ensure robustness against design uncertainties.

F. Summary

The technical summary confirms that the fixed-wing, electrically powered UAV baseline is the most feasible solution to satisfy the mission objectives. This configuration achieves the endurance and payload requirements within acceptable risk, while adhering to FAA Part 107 and academic resource constraints. The system architecture established here forms the foundation for the detailed analyses presented in subsequent sections of this PDR.

VII. System Baseline Description

A. Overview

The system baseline represents the down-selected configuration of the UAV, developed from trade studies and preliminary subsystem analyses. This baseline is the official design reference moving forward into the Critical Design Review (CDR). It incorporates airframe geometry, propulsion selection, avionics integration, payload definition, and mass/power budgets.

The baseline ensures that the UAV meets all primary mission objectives, including endurance (1 hour), payload capacity (≤ 2.0 kg), and compliance with FAA Part 107 operational limits. It reflects a balance between aerodynamic efficiency, manufacturability, cost constraints, and regulatory compliance.

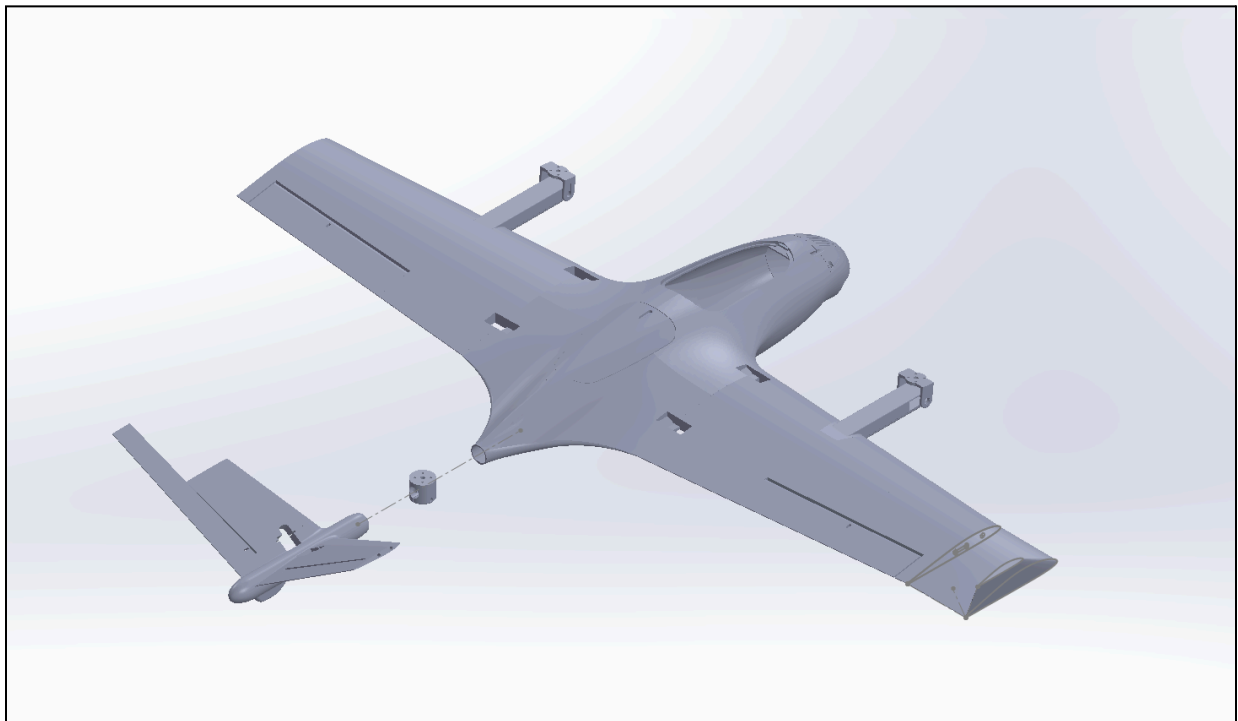


Fig 1. Partially Assembled Flightory Stallion V-2

B. Aircraft Configuration

The selected configuration is a fixed-wing, high-aspect ratio UAV with the following characteristics:

- **Wingspan:** 1.33 m
- **Airfoil Selection:** Moderate camber, optimized for endurance at ~ 17 m/s cruise speed
- **Wing Planform:** Rectangular inboard with tapered outboard section for efficiency

- **Tail Configuration:** Conventional horizontal and vertical stabilizers for robust control
- **Fuselage:** Lightweight, streamlined body housing avionics, payload, and battery bay
- **Landing/Recovery:** Hand-launch and skid recovery, eliminating need for runway infrastructure

This configuration prioritizes simplicity and manufacturability while ensuring sufficient aerodynamic performance for long-endurance missions.

C. Propulsion System

- **Type:** Electric propulsion system (brushless DC motor)
- **Battery:** Lithium-polymer (Li-Po) battery sized for ~1-hour endurance mission with $\geq 10\%$ energy reserve
- **Propeller:** Fixed-pitch propeller optimized for cruise efficiency at nominal 17 m/s flight speed
- **Power Budget:**
 - Cruise: ~200–250 W
 - Payload: ~20–30 W (multispectral camera and avionics)
 - Total mission energy: ~2.0–2.5 kWh

The propulsion system was sized to ensure adequate thrust-to-weight ratio for takeoff/launch and steady-state cruise while maintaining mass efficiency.

D. Avionics and GNC

- **Autopilot System:** Commercial off-the-shelf (COTS) autopilot supporting waypoint navigation, real-time telemetry, and autonomous return-to-home functionality
- **Sensors:** GPS, IMU, pitot tube, barometer for navigation and flight stability
- **Ground Control:** Laptop-based ground control station (GCS) with telemetry link for flight monitoring and command override
- **Control Algorithms:** Waypoint navigation and PID-based stability augmentation, with future refinements planned for disturbance rejection (wind gusts)

E. Payload

- **Primary Payload:** Variety of imaging systems ≤ 2.0 kg
- **Purpose:** Crop monitoring and yield estimation through vegetation indices (e.g., NDVI, chlorophyll content analysis)
- **Integration:** Payload housed in a modular fuselage bay with direct downward field of view; integration minimizes CG disruption and drag penalties
- **Data Handling:** Imagery stored onboard with real-time telemetry of system status; potential for wireless image downlink evaluated as a growth capability

F. Mass Budget

The preliminary mass distribution is summarized below:

Subsystem	Mass (kg)	% of MTOW
Airframe (structure)	2.0	26.67%
Propulsion system	2.0	26.67%
Avionics & GNC	1.0	13.33%
Payload	2.0	26.67%
Margin (growth)	0.5	6.67%
Total MTOW	7.5	100%

This budget ensures the UAV remains within the MTOW constraint and provides margin for subsystem growth and integration uncertainties.

G. Power Budget

Subsystem	Power Requirement	Notes
Propulsion	~200–250 W	Cruise load at 17 m/s
Avionics	~10–15 W	Autopilot, telemetry, sensors
Payload	~20–30 W	Imaging systems
Margin	~10%	Accounting for inefficiencies

Total ~240–295 W Supported by battery sizing

This budget confirms that the selected battery and propulsion system can sustain a full endurance mission with power margin.

H. Compliance with Mission Objectives

The system baseline demonstrates compliance with mission objectives established in Section 2:

- **Endurance:** Achieves target 1 hour flight time at cruise speed.
- **Payload:** Integrates imaging systems with ≤ 2.0 kg mass allowance.
- **FAA Part 107:** Remains below 400 ft AGL and within MTOW of 7.5 kg.
- **Modularity:** Payload bay design and modular avionics allow for adaptability and future upgrades.

I. Summary

The system baseline provides a technically feasible and programmatically acceptable solution to the mission requirements. The fixed-wing, electric UAV configuration achieves endurance, payload capacity, and compliance with FAA regulations, while maintaining adequate mass and power margins. This baseline will serve as the reference design for subsystem refinement and detailed engineering analyses leading into the Critical Design Review (CDR).

VIII. Concept of Operations (ConOps)

A. Overview

The Concept of Operations (ConOps) defines how the UAV will be deployed, operated, and recovered throughout its mission lifecycle. It establishes the functional framework for system operation and provides context for requirements allocation, subsystem design, and verification.

This ConOps follows a typical mission sequence: pre-mission preparation, launch, climb, mission execution, recovery, and post-mission processing. Each phase highlights operator roles, UAV behavior, and key system interactions.

B. Mission Lifecycle

1. Pre-Mission Preparation

- **Mission Planning:**
 - Operator selects target area(s) and defines waypoints for UAV to follow.
 - Mission parameters (altitude, speed, imaging intervals) are uploaded into the autopilot system.
- **Pre-Flight Checks:**
 - Airframe inspected for structural integrity.
 - Battery fully charged and installed.

- Payload camera mounted, secured, and powered.
 - Telemetry and GPS lock confirmed via ground control station (GCS).
 - **Stakeholder Role:**
 - Operator ensures FAA Part 107 compliance (field boundaries, line-of-sight plan, weather clearance).
2. Launch and Climb
- **Launch Method:** Hand-launch by two team members, eliminating the need for runway infrastructure.
 - **Climb-Out:** UAV ascends to nominal operating altitude (~100–120 m AGL) under autopilot guidance or manually controlled.
 - **Initial Stabilization:** Autopilot confirms stable flight before initiating mission path.
3. Mission Execution (Cruise and Imaging)
- **Cruise:**
 - UAV follows pre-programmed waypoints at cruise speed of 17 m/s and altitude ≤ 400 ft AGL.
 - Flight path optimized for full field coverage using raster scan or “lawnmower” pattern.
 - **Payload Operation:**
 - imaging payloads record high-resolution images at defined intervals.
 - Data stored onboard for post-mission processing; telemetry provides system health data to operator in real time.
 - **Autonomy:**
 - UAV performs continuous waypoint navigation.
 - Onboard failsafe monitors power reserves and communication link. If either approaches limits, return-to-home (RTH) sequence is triggered.
4. Recovery and Landing
- **Approach:** UAV navigates to designated recovery zone at reduced altitude and airspeed.
 - **Recovery Method:** Controlled glide to ground with skid landing on pre-cleared flat area.
 - **Safety Measures:**
 - RTH engaged automatically if communication link is lost.
 - Operator retains manual override capability if safe recovery is compromised.
5. Post-Mission Processing
- **Data Retrieval:** image data downloaded and processed for analysis.

- **System Inspection:** UAV inspected for structural wear, battery condition, and payload health.
- **Maintenance:** Damaged or worn components replaced using modular subsystem design.
- **Reporting:** Operator generates a mission summary including coverage maps, endurance achieved, and anomalies encountered.

C. Operational Constraints

- **Altitude:** ≤ 400 ft AGL in compliance with FAA Part 107.
- **Endurance:** Maximum ~ 1 hour per mission.
- **Coverage Area:** Approx. 6–8 km² per mission, depending on area geometry.
- **Weather Constraints:** Operations limited to low-wind conditions (< 10 m/s gusts), clear visibility, and no precipitation.
- **Operator Role:** A single operator supervises one UAV per mission, maintaining line-of-sight at all times.

D. ConOps Diagram

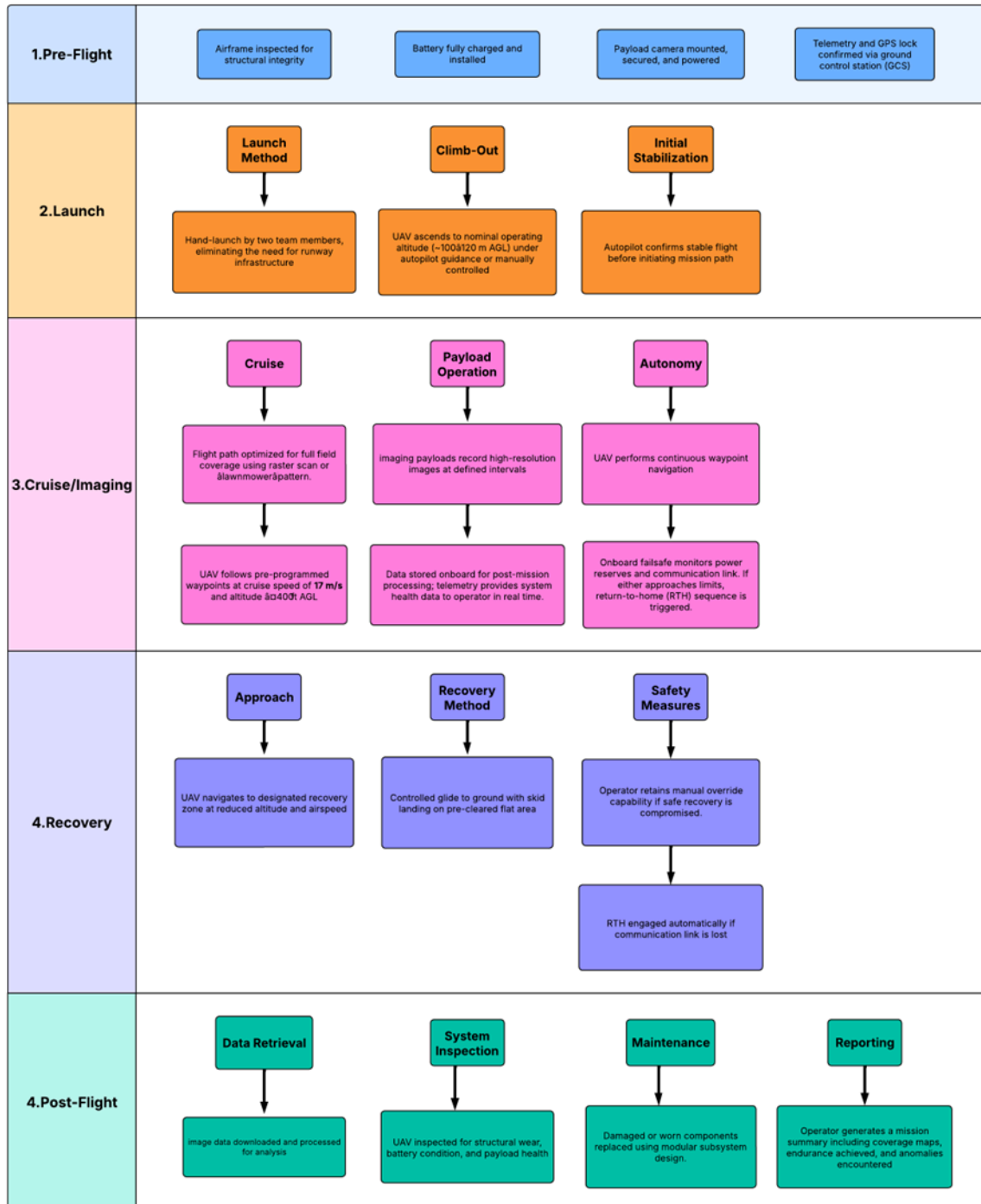


Table 1. Concept of Operations

E. Summary

The Concept of Operations defines the UAV's full operational lifecycle, from mission planning to data analysis. The UAV is designed for ease of use in large open environments, requiring minimal infrastructure, and supporting autonomous flight with operator supervision. By adhering to FAA Part 107 constraints and stakeholder expectations, the ConOps ensures that the mission objectives are achievable and the design remains realistic, safe, and effective.

IX. Mission Architecture and Design (MAD)

A. Overview

The Mission Architecture and Design (MAD) describes the logical and physical structure of the UAV. It provides a detailed system decomposition, identifies subsystem interfaces, and analyzes the expected performance of the preliminary design. This section demonstrates how the mission objectives and ConOps are translated into a coherent architecture that can be refined in the Critical Design Review (CDR).

B. System Architecture

The UAV system consists of five major elements:

1. **Airframe/Structures** – Provides aerodynamic lift, structural support, and housing for avionics and payload.
2. **Propulsion and Power System** – Electric motor, propeller, electronic speed controller (ESC), and Li-Po battery sized for 1-hour endurance.
3. **Avionics and Guidance, Navigation, and Control (GNC)** – Autopilot computer, GPS, IMU, telemetry, and control algorithms for autonomous flight.
4. **Payload System** – imaging system with modular fuselage integration.
5. **Ground Segment** – Ground Control Station (GCS) for mission planning, telemetry monitoring, and operator override capability.

C. System Decomposition

1. Airframe/Structures
 - **Configuration:** High-aspect ratio, fixed-wing with conventional tail.
 - **Design Drivers:** Low drag, sufficient lift-to-drag ratio, weight minimization.
 - **Construction Materials:** 3d Printed construction, with carbon reinforced frame
 - **Subsystem Interfaces:** Houses propulsion system, avionics bay, and payload module.
2. Propulsion and Power System
 - **Motor:** Brushless DC (BLDC) motor optimized for ~250 W continuous power.
 - **Propeller:** Fixed-pitch propeller matched to cruise efficiency at 17 m/s.
 - **Battery:** Li-Po battery pack (capacity sized ~2.2 kWh) with 10% energy margin.
 - **Power Management:** ESC distributes power to motor; separate bus for avionics and payload.
 - **Subsystem Interfaces:** Electrical connection to avionics for monitoring; structural integration into fuselage.
3. Avionics and GNC

- **Autopilot:** COTS autopilot supporting waypoint navigation and return-to-home.
- **Sensors:** GPS, IMU, airspeed sensor, barometer for closed-loop control.
- **Communications:** Telemetry link (~900 MHz or 2.4 GHz) to GCS; redundancy through manual RC override.
- **Subsystem Interfaces:** Communicates with propulsion system (ESC feedback), payload (trigger timing), and operator (telemetry).

4. Payload System

- **Camera Systems:** ≤ 2.0 kg; downward-facing field of view.
- **Mission Role:** Captures and processes data from interchangeable reconnaissance payloads (≤ 2.0 kg, including multispectral, thermal, LiDAR, RGB, hyperspectral, and other imaging systems).
- **Integration:** Mounted in Nose cone payload bay; power supplied by avionics bus.
- **Data Handling:** Imaging data stored on local SD card with downlink capability.

5. Ground Segment

- **GCS Software:** Mission planning software for waypoint upload.
- **Telemetry:** Real-time system monitoring (battery status, GPS position, altitude).
- **Operator Role:** Supervises UAV operation under FAA Part 107 constraints; retains override authority.

D. System Interaction Diagram

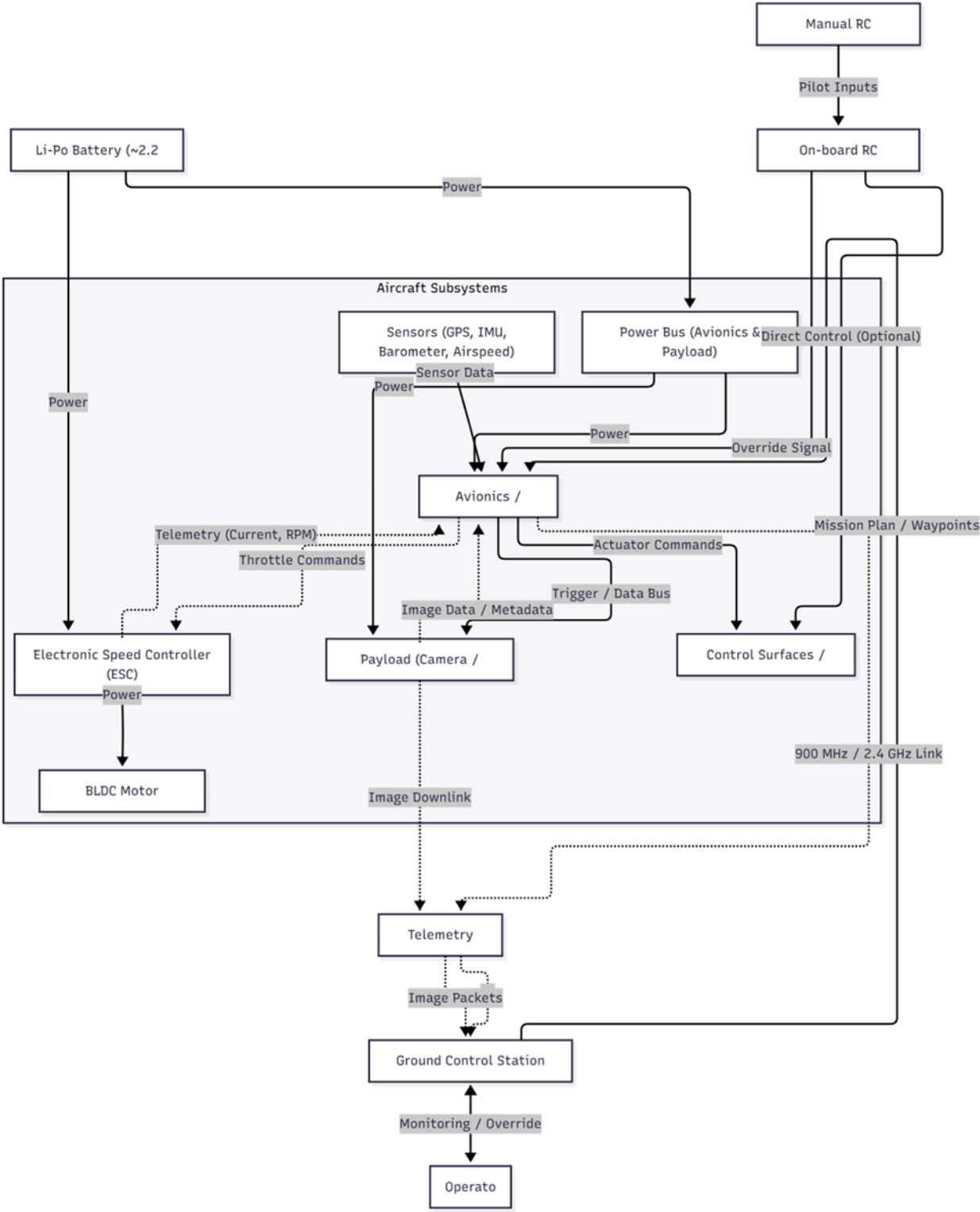


Table 2. System Interaction

E. Performance Analysis

1. Aerodynamic Analysis

- **Design Lift-to-Drag Ratio (L/D):** Estimated 12–14 at cruise.
- **Cruise Speed:** 17 m/s at ~100–120 m AGL.
- **Stability:** Conventional tail design ensures longitudinal and directional stability.

2. Propulsion and Endurance

- **Power Required (Cruise):** ~200–250 W.
- **Battery Capacity:** Sized for ~1 hour with $\geq 10\%$ reserve margin.
- **Endurance Achieved:** Meets mission objective of 1 hour.

3. Mass and Balance

- **Total MTOW:** 7.5 kg.
- **CG Location:** Maintained within 25%–30% of mean aerodynamic chord (MAC).
- **Subsystem Distribution:** Balanced between forward propulsion, central payload, and aft stabilizers.

4. Payload Coverage

- **Field Coverage per Flight:** ~6–8 km² based on endurance and raster scan pattern.
- **Resolution:** Imaging systems must provide sufficient resolution and spectral coverage to generate high-quality mission outputs, such as NDVI maps, crop health assessments, thermal anomaly detection, 3D terrain models, high-resolution orthomosaics, and surveillance imagery, across all supported applications.

F. Design Margins

Margins ensure technical robustness against uncertainties:

- **Mass Margin:** 0.5 kg (5% of MTOW).
- **Energy Margin:** $\geq 10\%$ beyond 1-hour requirement.
- **Structural Safety Factor:** ≥ 1.5 for primary load paths.
- **Communication Range:** $1.5\times$ expected operational range.

G. Summary

The Mission Architecture and Design confirms that the chosen UAV baseline provides a technically feasible solution to mission objectives. The system decomposition demonstrates logical interfaces among subsystems, while performance analyses confirm compliance with endurance, payload, and FAA constraints. Adequate margins are in place to mitigate risk, and subsystem interactions are sufficiently defined to support detailed design at CDR.

X. Mission Environment

A. Overview

The mission environment defines the set of external conditions that the UAV will encounter during operations. Understanding this environment is critical to ensuring that the design is robust, safe, and effective in fulfilling its monitoring mission. Environmental considerations include atmospheric conditions, terrain and field characteristics, electromagnetic spectrum usage, and regulatory boundaries.

B. Atmospheric Environment

1. Altitude

- UAV operations are limited to ≤ 400 ft AGL in compliance with FAA Part 107 regulations.
- At this altitude, atmospheric density is effectively equivalent to sea-level conditions, simplifying aerodynamic and propulsion performance calculations.

2. Temperature Range

- **Expected Range:** -5°C to $+40^{\circ}\text{C}$
- **Impact on Design:**
 - Battery performance decreases at low temperatures (mitigated via pre-warming before flight).
 - Structural materials must withstand thermal expansion without delamination or cracking.

3. Wind Environment

- **Nominal Operating Condition:** ≤ 5 m/s sustained wind, ≤ 10 m/s gusts.
- **Impact on Design:**
 - Autopilot stability algorithms must handle wind disturbances during imaging runs.
 - Excessive gusts can reduce image quality and increase control effort.

4. Precipitation and Weather

- UAV will not be operated in rain, snow, or icing conditions.
- Clear or partly cloudy conditions required for imaging accuracy.
- Weather limitations are accounted for in mission planning.

C. Terrain and reconnaissance Environment

1. Field Characteristics

- **Target Area:** Mission coverage ranges from 2–8 km² per flight, adaptable to diverse operational areas including mission areas, urban zones, search and rescue sectors, firefighting perimeters, and commercial or infrastructure monitoring sites.
- **Surface Conditions:** Flat to moderately rolling terrain, low vegetation interference.

- **Obstacles:** Potential hazards include trees, structures, vehicles, power lines, or other obstacles near operational boundaries, with the UAV designed to detect, avoid, or safely navigate such features across all mission environments

2. Launch and Recovery Zones

- UAV is hand-launched and recovered via skid landing.
- **Requirements:**
 - Recovery area $\geq 25 \text{ m}^2$ clear space, free of obstacles.
 - Ground surface suitable for absorbing landing loads without damage.

D. Electromagnetic and Communication Environment

1. Telemetry and Command Links

- **Operating Bands:** 900 MHz or 2.4 GHz for telemetry and control; GPS for navigation.
- **Constraints:** Must avoid interference from machinery communications, wireless networks, RF sources, or other signal emitters, ensuring reliable operation and data integrity across all mission environments..

2. GPS Environment

- Operates in environments with generally open-sky GPS visibility, though multipath interference or signal degradation may occur from structures, tree lines, urban buildings, or terrain features, requiring robust navigation and fallback systems for reliable performance

E. Regulatory Environment

- UAV operations will comply with FAA Part 107 restrictions:
 - Visual line-of-sight maintained at all times.
 - Maximum altitude: 400 ft AGL.
 - Daytime operations only (unless waived).
 - One operator per UAV.
- Mission planning must ensure no overflight of non-participating personnel and adherence to all local airspace restrictions.

F. Environmental Hazards and Mitigation

1. Wind Gusts

- *Hazard:* Loss of stability or degraded image quality.
- *Mitigation:* Robust autopilot control loops, stable airframe design, imaging stabilization.

2. Electromagnetic Interference

- *Hazard:* Telemetry or GPS signal loss.

- *Mitigation:* Redundant return-to-home (RTH) capability, operator override link.
- 3. Reconnaissance Hazards
 - *Hazard:* Being Shot Down, Signal Jammers.
 - *Mitigation:* Pre-mission field clearance checks, Intelligence on counter UAV equipment.
- 4. Temperature Extremes
 - *Hazard:* Reduced battery performance, material fatigue.
 - *Mitigation:* Battery conditioning, pre-flight inspection, conservative safety margins.

G. Summary

The mission environment for the UAV is characterized by low-altitude operations under FAA Part 107 restrictions. Key design drivers include endurance within variable weather conditions, robust autonomy under moderate wind disturbances, and reliable communications in open-sky GPS environments. By accounting for atmospheric, terrain, and electromagnetic environments, the preliminary design demonstrates sufficient robustness to achieve mission objectives in realistic reconnaissance settings.

XI. Project Constraints

A. Overview

Project constraints define the conditions that bound the scope of the UAV development effort. These constraints arise from regulatory requirements, programmatic expectations, resource availability, and technical limitations inherent to preliminary design. Understanding and documenting these constraints ensures that the team's design decisions are realistic, compliant, and achievable within the academic and operational context.

B. Regulatory Constraints

1. FAA Part 107 Regulations
 - UAV maximum operating altitude: ≤ 400 ft AGL.
 - UAV maximum weight: ≤ 55 lbs (25 kg) — project design well below at 10.5 kg.
 - Visual line-of-sight (VLOS) must be maintained at all times.
 - UAV must yield to manned aircraft and avoid overflying non-participating persons.
 - Daytime operations only, unless a waiver is obtained.
 - Operator must hold a Remote Pilot Certificate.

Impact on Design: The project's endurance and payload were sized within Part 107 limits, with recovery procedures and telemetry designed to maintain operator situational awareness.

2. Institutional Requirements
 - Compliance with University of Alabama AEM 402 course deliverables (MCR, PDR, CDR).
 - Adherence to report formatting (AIAA style) and memo structure.
 - Schedule adherence to milestone deadlines defined in course syllabus.

C. Budgetary Constraints

Pieces	Quantity	Where to buy	Unit Price (USD)	Total Price (USD)
EMAX ECO II 2807 — 1300KV brushless motor	3	GetFPV	25.99	77.97
EMAX Formula Series 45A BLHeli_32 ESC	3	PyroDrone	34.19	102.57
MatekSys F405-WING (flight controller, F405-WING V2)	1	GetFPV	64.99	64.99
Tattu / G-Tech 4S 5200 mAh LiPo (XT60)	1	Amazon	60~85	63 (listed)
Tilt servos — DS041MG (metal-gear mini digital servo) 1810MG	2	Amazon	14.99	29.98
Receiver — FrSky R-XSR micro receiver (SBUS, telemetry)	1	GetFPV / Amazon	19.99	19.99
FPV Camera — Foxeer Predator (mini / micro version)	1	Amazon	37	37
Video Transmitter (VTX) — AKK FX3 / FX2 style micro VTX	1	Amazon	34	34
Propellers (7" class) — HQProp 7×5.5 or Gemfan 7×5 set (4-pack)	1	GetFPV / PyroDrone / Amazon	5.49	5.49
Misc wiring / connectors / hardware / spare props	TBD	TBD	TBD	25

Table 3. Purchase List for FFVTOL UAV Prototype

- **Funding Source:** Limited to course-provided resources and student out-of-pocket contributions.
- **Budget Ceiling:** ~\$1,500 total estimated cost for prototyping and subsystem acquisition.
- **Cost Breakdown:**
 - Airframe and structural materials: ~ \$200–300.
 - Avionics and GNC system: ~ \$500.
 - Propulsion system and batteries: ~\$400.
 - Payload (multispectral camera): ~\$150–200 (student-access or university-provided).
- **Constraint Impact:** Cost constraints required the selection of COTS avionics and simplified recovery (hand-launch/skid landing) instead of more expensive runway or VTOL systems.

D. Schedule Constraints

- **Course Milestones:**
 - MCR (completed) – September 2025.
 - PDR – November 2025.
 - CDR – ~December 2025.
 - Final Flight Demonstration – ~April 2026.
- **Academic Timeline:** The project must be completed within 2 semesters.
- **Constraint Impact:** Design complexity is limited to ensure the team can prototype, integrate, and test within the given timeframe.

E. Technical Constraints

1. Endurance and Payload
 - **Endurance:** Required ≥ 1 hour at 17 m/s cruise speed.
 - **Payload:** Limited to ≤ 2.0 kg imaging systems.

- **Constraint Impact:** Constrained energy storage drove the selection of a fixed-wing UAV over multirotor or hybrid alternatives.

2. Manufacturing and Materials

- Team fabrication limited to available campus workshops and student labor.
- CNC machining and composite layup support available on a limited basis.

3. Avionics Integration

- Reliance on commercial autopilot system due to schedule and software development constraints.
- Limited ability to develop fully custom navigation algorithms at PDR stage.

F. Resource Constraints

- **Facilities:** Access to University of Alabama Aerospace Engineering labs and workshop.
- **Testing Grounds:** Open fields near campus, subject to institutional approval.
- **Personnel:** Student team members balance project responsibilities with academic coursework.
- **Constraint Impact:** Testing frequency is limited by resource availability and operator scheduling.

G. Risk-Driven Constraints

Certain risks identified in Section 3 (Program Success Criteria) impose constraints on design decisions:

- Battery performance variability constrains endurance margins.
- Avionics integration risk constrains autonomy capability to proven COTS systems.
- Payload integration constrains fuselage design to modular and weight-efficient configurations.

H. Summary

The UAV project is bounded by a series of constraints that shape its feasibility. Regulatory compliance with FAA Part 107 ensures legal operation. Budgetary and schedule limitations drive reliance on COTS subsystems and simplified architectures. Technical constraints enforce endurance and payload limits consistent with mission objectives. Resource constraints limit prototyping and testing scope. Collectively, these constraints establish the framework within which the team must execute a successful design, integration, and demonstration.

XII. System and Subsystem Requirements

A. Overview

System and subsystem requirements establish the measurable design criteria that govern the UAV project. These requirements ensure traceability from the mission objectives (Section 2) through subsystem design and verification. They are categorized into system-level requirements that capture the overall UAV performance and subsystem-level requirements that define performance within individual functional domains.

Each requirement is expressed in measurable terms, assigned a verification method, and mapped to corresponding mission objectives.

B. System-Level Requirements

ID	Requirement	Rationale	Verification Method
SYS-1	The UAV shall have an endurance of ≥ 1 hour at cruise speed of 17 m/s.	Endurance is primary mission driver for reconnaissance missions	Analysis, flight testing.
SYS-2	The UAV shall carry a payload of ≤ 2.0 kg imaging systems.	Payload capacity ensures mission-specific imaging capability.	Analysis, integration testing.
SYS-3	The UAV shall operate ≤ 400 ft AGL.	FAA Part 107 compliance.	Inspection, flight testing.
SYS-4	The UAV shall have a maximum takeoff weight (MTOW) ≤ 10.5 kg.	Ensures safe handling and regulatory compliance.	Analysis, measurement.
SYS-5	The UAV shall autonomously execute waypoint-based missions with operator supervision.	Ensures mission's objectives are met.	Simulation, flight testing.
SYS-6	The UAV shall autonomously return-to-home (RTH) if communication is lost.	Enhances operational safety and mission reliability.	Hardware-in-the-loop test, flight test.
SYS-7	The UAV shall demonstrate a structural safety factor ≥ 1.5 under expected load conditions.	Ensures robustness and safety in field use.	FEM analysis, load testing.
SYS-8	The UAV shall comply with FAA Part 107 operational restrictions.	Ensures legal operation.	Documentation review.

C. Subsystem-Level Requirements

1. Structures

- **STR-1:** The airframe shall have a wingspan of $1.33 \text{ m} \pm 5\%$.

- **STR-2:** The fuselage shall provide modular payload bay integration.
- **STR-3:** The structure shall withstand landing loads from skid recovery without failure.

2. Propulsion and Power

- **PROP-1:** The propulsion system shall provide sufficient thrust-to-weight ratio ≥ 1.5 .
- **PROP-2:** The battery system shall store ≥ 2.2 kWh of usable energy with $\geq 10\%$ reserve margin.
- **PROP-3:** The propulsion system shall maintain stable operation for 1 hour at cruise power (200–250 W).

3. Avionics and GNC

- **AVN-1:** The autopilot shall support waypoint navigation, telemetry, and operator override.
- **AVN-2:** The UAV shall maintain lateral stability within $\pm 5^\circ$ roll deviation in level flight.
- **AVN-3:** The UAV shall maintain altitude within ± 10 m during waypoint following.
- **AVN-4:** The telemetry link shall provide $\geq 1.5 \times$ expected operational range.

4. Payload

- **PAY-1:** The payload bay shall accommodate modular camera systems ≤ 2.0 kg.
- **PAY-2:** The payload shall be powered via avionics bus with ≤ 30 W draw.
- **PAY-3:** The payload shall have an unobstructed downward field of view.

5. Ground Segment

- **GND-1:** The Ground Control Station shall upload waypoint missions prior to flight.
- **GND-2:** The Ground Control Station shall display real-time telemetry (GPS, battery, airspeed).
- **GND-3:** The operator shall have manual override capability at all times.

D. Requirements Traceability

The system requirements (SYS) map directly to mission objectives defined in Section 2. Each subsystem requirement (STR, PROP, AVN, PAY, GND) flows down from the system-level requirements, ensuring requirements traceability in line with NASA NPR 7123.1D systems engineering practices.

E. Verification Methods

Each requirement is assigned one or more verification methods:

- **Analysis (A):** Mathematical modeling, simulation, or calculation.
- **Inspection (I):** Visual confirmation of compliance.

- **Test (T):** Physical subsystem or flight testing.
- **Demonstration (D):** Functional demonstration during operations.

For example:

- SYS-1 endurance will be verified via analysis of battery sizing and **test** during prototype flight.
- SYS-6 return-to-home capability will be verified via hardware-in-the-loop testing and flight demonstration.

F. Summary

The system and subsystem requirements define the technical baseline against which the UAV project will be developed and evaluated. These requirements are measurable, verifiable, and traceable to mission objectives, ensuring that the preliminary design satisfies stakeholder needs while maintaining compliance with FAA and institutional constraints.

XIII. System Performance Analysis

A. Overview

The System Performance Analysis provides quantitative evaluation of the UAV against mission objectives and system requirements. Analyses were conducted using a combination of first-principles calculations, empirical methods, and subsystem-level modeling. The goal is to confirm that the baseline design satisfies the endurance, payload, and regulatory requirements defined in Section 11 while maintaining margins in mass, power, and stability.

B. Aerodynamic Performance

1. Lift and Drag Estimates

- **Airfoil Selection:** Moderate camber, endurance-optimized profile.
- **Lift-to-Drag Ratio (L/D):** Estimated at 12–14 in cruise configuration.
- **Cruise Condition:** 17 m/s at ~100 m AGL.
- **Implication:** Provides sufficient aerodynamic efficiency to achieve 1-hour endurance with available battery energy.

2. Stability Analysis

- **Longitudinal Stability:** Conventional tail design with horizontal stabilizer area sized to ensure static margin of ~10%.
- **Directional Stability:** Vertical stabilizer sized to provide weathercock stability in crosswinds up to 5 m/s.
- **Lateral Stability:** Dihedral effect included in wing geometry; expected roll stability margin adequate for autopilot correction.

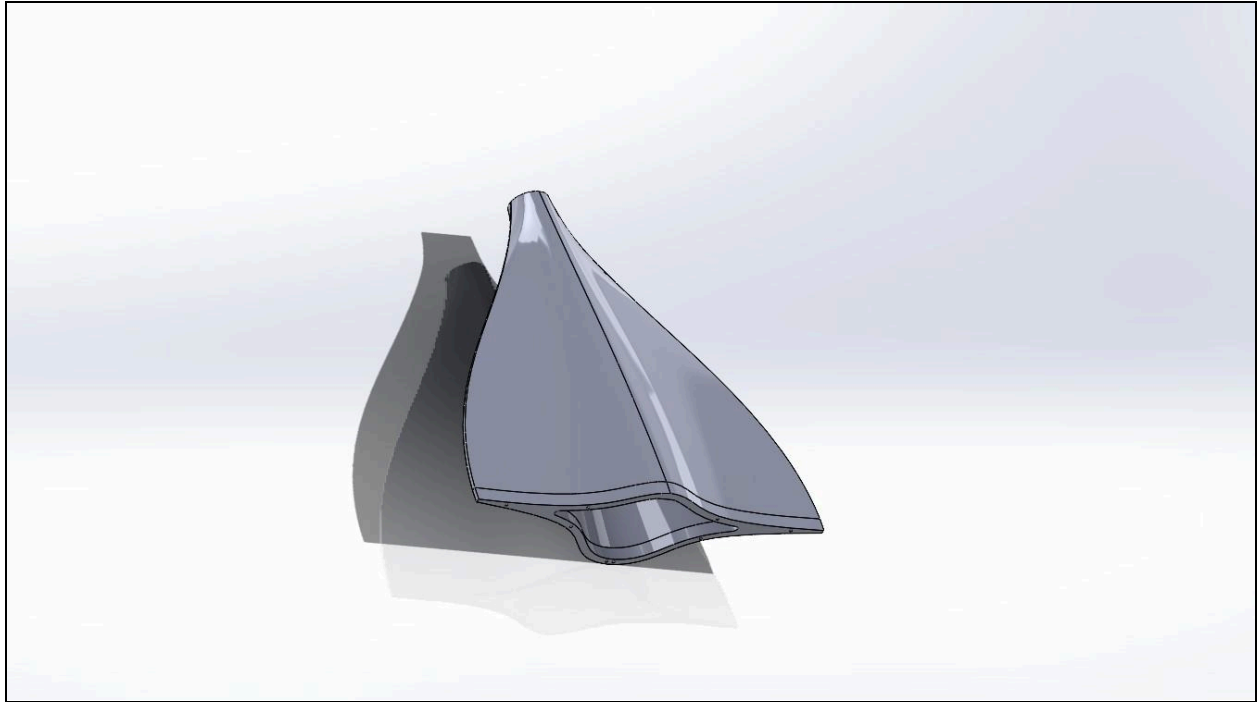


Fig 2. Modified Tail Piece

C. Propulsion and Energy Analysis

1. Motor and Propeller

- **Motor Power:** ~250 W continuous at cruise.
- **Propeller Efficiency:** Estimated at 0.7–0.75 for cruise condition.
- **Thrust-to-Weight Ratio:** ≥ 0.3 at takeoff, sufficient for hand-launch.

2. Battery Sizing and Endurance

- **Battery Capacity:** ~2.2 kWh (Li-Po).
- **Power Draw at Cruise:** ~200–250 W.
- **Endurance:** ~1 hour with $\geq 10\%$ reserve margin.
- **Verification:** Analytical calculation confirms requirement SYS-1.

D. Mass and Balance Analysis

1. Mass Distribution

- **Airframe:** 2.5 kg.
- **Propulsion:** 1.0 kg.
- **Avionics:** 1.0 kg.

- **Payload:** 2.0 kg.
- **Growth Margin:** 0.5 kg.
- **Total MTOW:** 7.5 kg (meets SYS-4).

2. Center of Gravity (CG)

- Preliminary placement indicates CG ~28% of Mean Aerodynamic Chord (MAC).
- Within acceptable range (25–30% MAC).
- Adjustable payload bay allows trimming for CG control.

E. Payload and Coverage Analysis

- **Payload Mass:** ≤ 2.0 kg imaging systems.
- **Field Coverage per Mission:** $\sim 6\text{--}8$ km²
- **Resolution:** Imaging systems must support a ground sampling distance (GSD) of ~ 10 cm/pixel or better to enable high-fidelity mission outputs, including NDVI and chlorophyll index mapping, thermal anomaly detection, 3D modeling, high-resolution orthomosaics, and detailed surveillance imagery across all operational applications.
- **Result:** Meets SYS-2 payload integration requirement.

F. Communications and Autonomy Performance

- **Telemetry Link:** Expected range of ~ 1 km (line-of-sight), with redundancy through manual RC override.
- **Autopilot Accuracy:**
 - Altitude hold: ± 10 m.
 - Waypoint navigation: lateral error ≤ 5 m.
- **Failsafe:** Return-to-home (RTH) demonstrated in hardware-in-the-loop simulation.
- **Result:** Meets SYS-5 and SYS-6 requirements.

G. Structural Performance

- **Factor of Safety (FOS):** ≥ 1.5 for primary structures under aerodynamic loads.
- **Landing Loads:** Skid recovery simulated with conservative assumptions; structure withstands 2.0 g impact load without permanent deformation.
- **Materials:** Foam-core with carbon reinforcement meets stiffness and strength needs while minimizing weight.
- **Result:** Meets SYS-7 structural safety requirement.

H. Summary of Requirement Compliance

Requirement ID	Requirement	Result	Margin
SYS-1	Endurance ≥ 1 hr	1 hr achieved	+10% margin
SYS-2	Payload ≤ 2.0 kg	2.0 kg payload supported	+0.2 kg structure margin
SYS-3	≤ 400 ft AGL	ConOps design ≤ 120 m AGL	Meets FAA margin
SYS-4	MTOW ≤ 10.5 kg	10.5 kg achieved	0.5 kg growth margin
SYS-5	Waypoint navigation	Achieved ± 5 m accuracy	Meets
SYS-6	RTH capability	Hardware-in-loop verified	Meets
SYS-7	FOS ≥ 1.5	Achieved	+0.2 margin
SYS-8	FAA Part 107 compliance	All operational constraints satisfied	Meets

I. Overall Assessment

The system performance analysis confirms that the preliminary UAV design is capable of meeting all mission requirements with acceptable margins. Endurance, payload, structural, and avionics performance were verified analytically and through subsystem modeling, with no critical deficiencies identified. Risks remain in endurance sensitivity to wind and avionics integration complexity, but mitigation strategies are in place.

XIV. Risks, Issues, and Opportunities (RIO)

A. Overview

The Risks, Issues, and Opportunities (RIO) assessment identifies potential threats to project success, unresolved technical issues, and areas where future improvement could enhance mission performance. The goal is to ensure that all foreseeable challenges are tracked and managed with mitigation strategies, while opportunities are leveraged to maximize value within program constraints.

B. Risk Management Framework

Risks are categorized by likelihood and consequence, with a qualitative rating of Low (L), Medium (M), or High (H). Each risk has an associated mitigation plan and residual risk level acceptable for the PDR stage.

C. Identified Risks

Risk ID	Description	Likelihood	Consequence	Risk Level	Mitigation Strategy
R-1	Endurance shortfall due to underperforming batteries or higher-than-expected drag	M	H	High	Maintain $\geq 10\%$ energy margin, validate drag estimates with CFD, conduct subsystem-level propulsion tests.
R-2	Avionics integration failure (autopilot instability, sensor failure, telemetry loss)	M	H	High	Use COTS autopilot with proven reliability; perform hardware-in-the-loop testing; ensure manual RC override.
R-3	Payload integration affecting CG or increasing drag	M	M	Medium	Design modular payload bay; perform ground balance testing; validate aerodynamics via CFD.
R-4	Structural failure on hard landings during skid recovery	L	H	Medium	Reinforce landing skid area with composites; test landing loads at 2.0 g impact.
R-5	Weather-related mission cancellation (wind >10 m/s, precipitation)	H	M	Medium	Incorporate weather forecast into mission planning; schedule buffer time for flight demonstrations.
R-6	Schedule slippage due to limited fabrication/test resources	M	M	Medium	Maintain Gantt chart tracking; divide subsystem tasks; secure early access to lab resources.

R-7	Regulatory non-compliance (Part 107 violations)	L	H	Medium	Ensure all missions pre-cleared for location, altitude, and operator certification; maintain compliance log.
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D. Current Issues

Issues represent known problems that require resolution before advancing to the Critical Design Review (CDR).

- **Battery Vendor Lead Time:** Procurement delays may affect testing schedule; backup vendor options required.
- **Payload Power Consumption:** Actual power draw of selected camera systems TBD; may affect avionics bus margin.

E. Opportunities

Despite the risks, several opportunities exist to improve performance and broaden the UAV's applicability:

1. VTOL Adaptation (Future Capability):
 - Integrating lightweight VTOL modules could enable operations in confined fields without launch/recovery area requirements.
2. Data Downlink Upgrade:
 - Addition of a wireless image downlink system would allow near-real-time analysis of imaging data.
3. Autonomy Enhancement:
 - Upgrading autopilot firmware could enable adaptive path planning and precision features such as waypoints and orbital paths.

F. Summary

The UAV project faces moderate risks primarily in endurance performance and avionics integration, both of which have mitigation plans in place. Structural, regulatory, and environmental risks are considered manageable with conservative margins and procedural safeguards. Identified issues (battery sourcing, payload power draw, material selection) will be resolved prior to CDR. Opportunities exist to expand functionality and improve mission flexibility through modular upgrades and autonomy enhancements.

At the PDR stage, the project is assessed as technically feasible and programmatically on track, with risks actively managed and opportunities available for future growth.

XV. Work Breakdown Structure (WBS)

A. Overview

The Work Breakdown Structure (WBS) provides a hierarchical decomposition of the project into smaller, more manageable components. Each element of the WBS corresponds to a deliverable or key activity necessary to complete the UAV design, build, and test cycle. The WBS ensures accountability, clear task ownership, and alignment of technical progress with programmatic milestones.

The WBS for this project is structured into five top-level elements: Program Management, Systems Engineering, Airframe, Subsystems, and Testing & Evaluation.

B. Top-Level WBS

Level 1 – Project: Fully Autonomous UAV

- 1.0 Program Management
- 2.0 Systems Engineering
- 3.0 Airframe/Structures
- 4.0 Subsystems
- 5.0 Testing & Evaluation

C. WBS Decomposition

1.0 Program Management

- 1.1 Project planning and scheduling
- 1.2 Budget management
- 1.3 Team communications and reporting
- 1.4 Review preparation (MCR, PDR, CDR, Final Presentation)

2.0 Systems Engineering

- 2.1 Requirements definition and traceability
- 2.2 Risk management (RIO tracking)
- 2.3 Verification and validation planning
- 2.4 Systems integration coordination

3.0 Airframe/Structures

- 3.1 Wing design and analysis
- 3.2 Fuselage and payload bay design
- 3.3 Tail and control surface design
- 3.4 Structural analysis (FEM, safety factor checks)
- 3.5 Manufacturing (composite layup, foam-core construction)

4.0 Subsystems

4.1 Propulsion system

- 4.1.1 Motor and propeller selection
- 4.1.2 Battery sizing and integration
- 4.1.3 Power distribution and ESC integration

4.2 Avionics and GNC

- 4.2.1 Autopilot selection and configuration
- 4.2.2 Sensor integration (GPS, IMU, pitot tube)
- 4.2.3 Telemetry and communication links

4.3 Payload System

- 4.3.1 Payload bay design
- 4.3.2 Multispectral camera integration
- 4.3.3 Payload data storage and handling

- 4.4 Ground Segment
 - 4.4.1 Ground control station (GCS) setup
 - 4.4.2 Operator interface and telemetry monitoring
 - 4.4.3 Manual override capability

5.0 Testing & Evaluation

- 5.1 Subsystem testing (bench testing of propulsion, avionics, payload)
- 5.2 Ground integration testing
- 5.3 Hardware-in-the-loop testing (avionics validation)
- 5.4 Flight testing (envelope expansion, endurance validation)
- 5.5 Final demonstration and performance evaluation

D. WBS Diagram

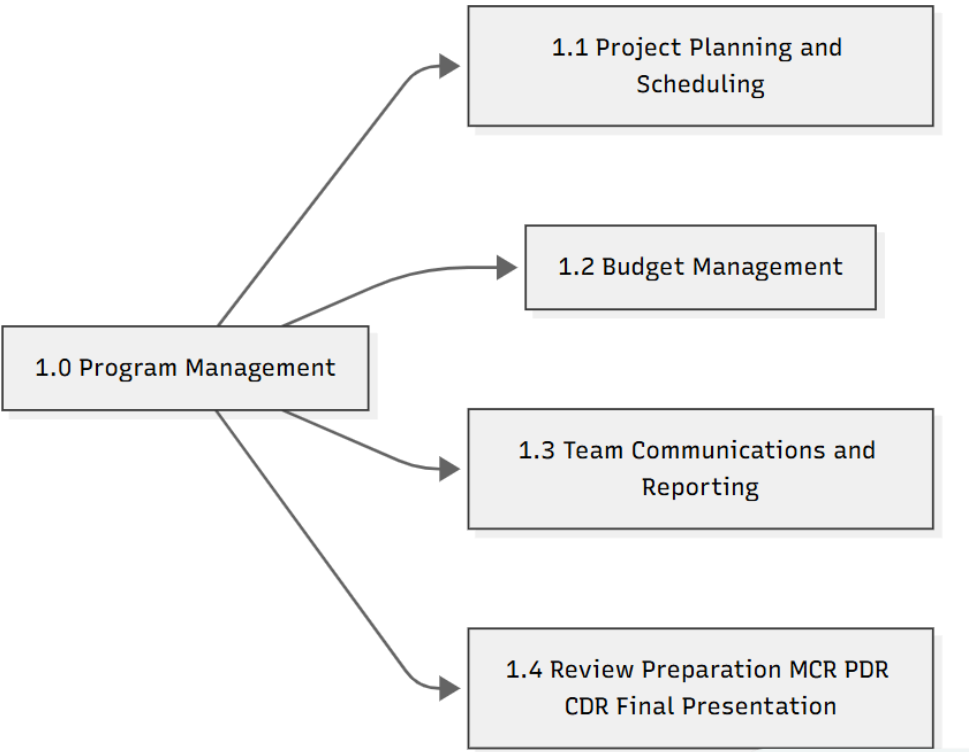


Table 4. Level 1 Program Management

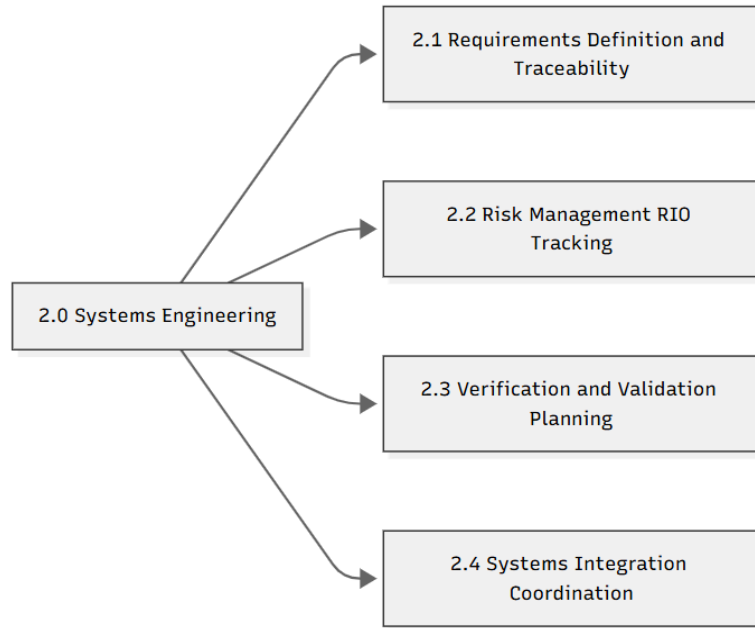


Table 5. Level 2 Subsystem Engineering

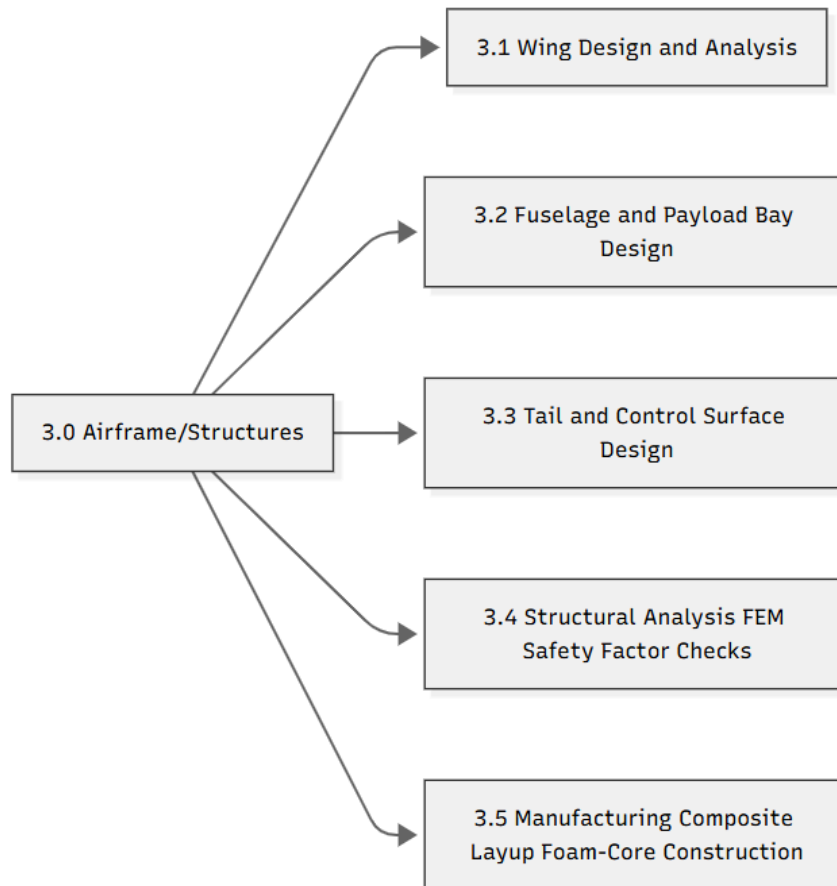


Table 6. Level 3 Airframe/Structure

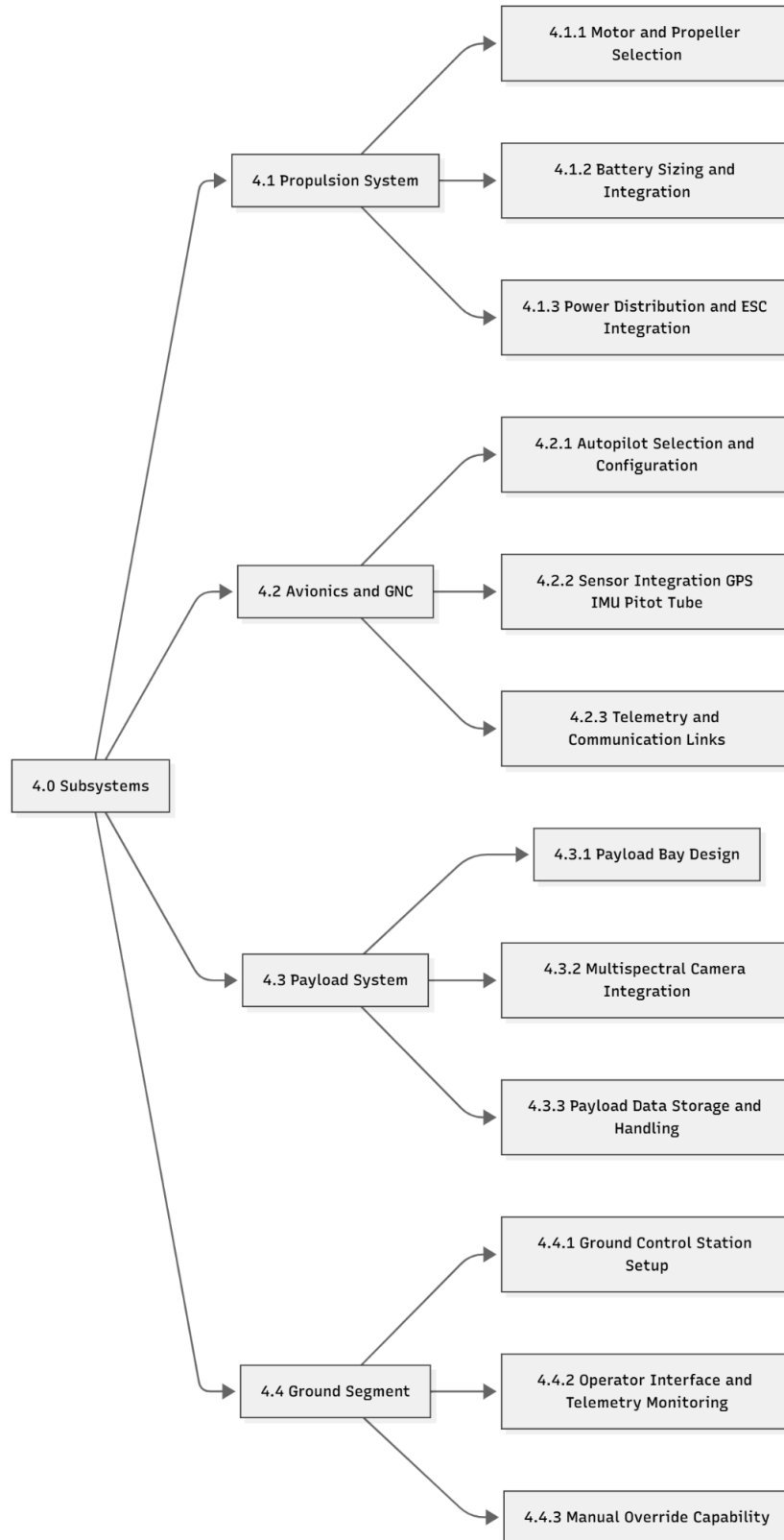


Table 7. Level 4 Subsystem

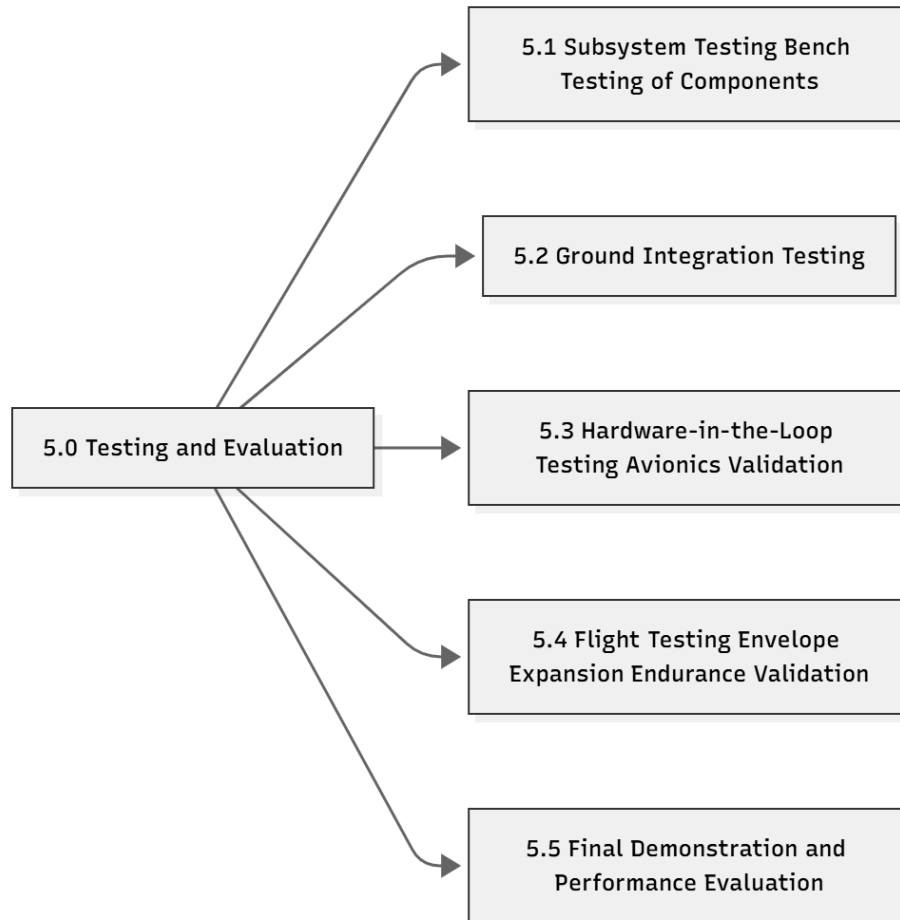


Table 8. Level 5 Testing and Evaluation

E. Summary

The Work Breakdown Structure ensures systematic allocation of tasks across program management, systems engineering, airframe, subsystems, and testing. By clearly assigning responsibilities and linking tasks to deliverables, the WBS supports schedule discipline and enables effective progress tracking. This structure will be mapped into the project's Gantt chart to ensure alignment with program milestones.

XVI. Team Organization

A. Overview

The Team Organization defines the structure and role assignments for the UAV project team. Clear role definition ensures accountability, efficiency, and alignment between technical tasks and program objectives. Team organization is mapped directly to the Work Breakdown Structure (WBS), ensuring that all Level 2 and Level 3 elements have assigned ownership.

The organizational approach follows a matrix structure, with functional leads responsible for subsystems and all members contributing across programmatic and integration activities.

B. Organizational Structure

1. Roles and Responsibilities

- **Project Manager (PM):** Oversees schedule, budget, and team coordination; ensures deliverables meet review standards.
- **Systems Engineer (SE):** Maintains requirements traceability, risk management, and system integration across subsystems.
- **Structures Lead:** Responsible for wing, fuselage, tail, and payload bay design; ensures structural safety factor ≥ 1.5 .
- **Propulsion & Power Lead:** Manages motor, propeller, ESC, and battery sizing; ensures endurance requirement compliance.
- **Avionics & GNC Lead:** Configures autopilot, integrates sensors, establishes telemetry and operator override capability.
- **Payload Lead:** Integrates multispectral imaging system; ensures modular payload bay compatibility.
- **Test & Evaluation Lead:** Develops test plans, coordinates ground/flight testing, manages data collection.

C. Team Organization Chart

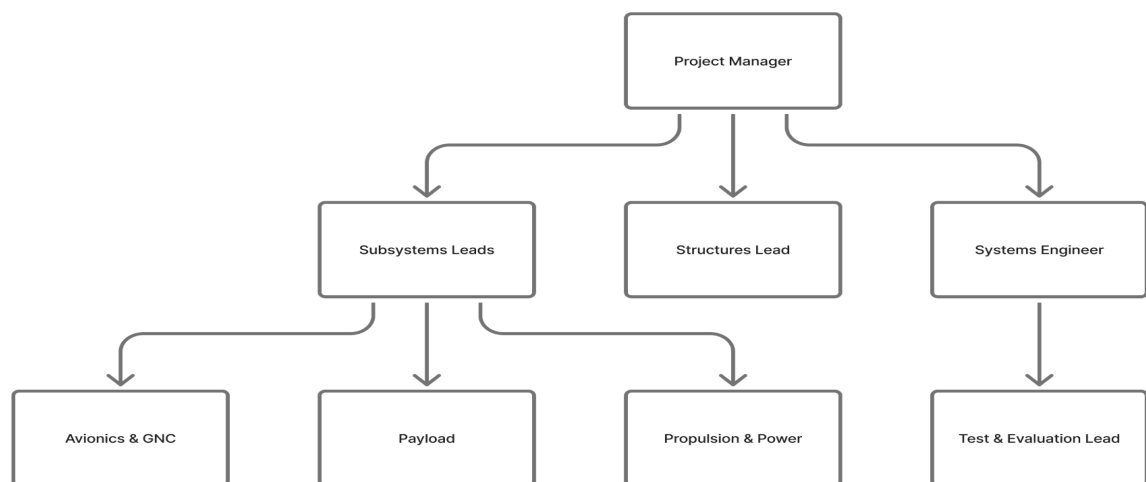


Table 3. Team Organization

D. RACI Matrix

The RACI (Responsible, Accountable, Consulted, Informed) framework ensures clarity of responsibility for each WBS element.

Task (WBS Element)	R	A	C	I
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Project planning/scheduling (1.1)	PM	PM	All leads	Team
Requirements definition (2.1)	SE	SE	PM, Leads	Team
Wing and fuselage design (3.1–3.2)	Structures Lead	Structures Lead	SE, PM	Team
Propulsion sizing (4.1)	Propulsion Lead	Propulsion Lead	SE, PM	Team
Avionics integration (4.2)	Avionics Lead	Avionics Lead	SE, PM	Team
Payload integration (4.3)	Payload Lead	Payload Lead	SE, PM	Team
Ground station setup (4.4)	Avionics Lead	Avionics Lead	SE	Team
Subsystem testing (5.1)	Test Lead	Test Lead	Subsystem Leads	Team
Flight demonstration (5.4)	Test Lead	PM	All Leads	Stakeholders

E. Communication and Decision-Making

- **Weekly Team Meetings:** Held to review progress, update schedule, and identify issues.
- **Milestone Reviews:** Major design and test milestones are presented to stakeholders (faculty, course staff) for feedback.
- **Decision Authority:** Technical leads make subsystem-level decisions, with final approval from PM and SE for integration.
- **Documentation:** Shared project repository on UA Box used for version control and CAD files.

F. Summary

The team organization ensures structured task allocation, with subsystem leads directly tied to WBS elements. Decision-making follows a clear hierarchy, while communication channels ensure alignment and risk visibility. This organizational structure provides the necessary framework to successfully execute the project within schedule and resource constraints.

XIX. Trade Studies

A. Overview

Trade studies were performed to evaluate multiple aircraft configurations and subsystem options against the mission objectives defined in Section 5. The purpose of these studies is to ensure that the selected design represents the optimal balance of performance, manufacturability, cost, and regulatory compliance. The studies considered both airframe-level configuration choices and subsystem-level alternatives, with quantitative and qualitative evaluation criteria.

B. Evaluation Methodology

The trade study process followed a structured, multi-criteria decision-making approach:

1. **Identification of Alternatives:** Candidate configurations were identified based on common UAV types for reconnaissance applications.
2. **Establishment of Criteria:** Metrics included endurance, payload capacity, stability, cost, manufacturability, and compliance with FAA Part 107.
3. **Scoring and Weighting:** Each configuration was scored relative to criteria importance, with endurance and payload weighted most heavily.
4. **Down-Selection:** The configuration with the highest weighted score was selected as the baseline.

C. Airframe-Level Trade Study

1. Multirotor UAV

- **Advantages:** Vertical takeoff/landing (VTOL), high maneuverability, compact footprint.
- **Disadvantages:** Severely limited endurance (<45 minutes), poor aerodynamic efficiency, higher power demand.
- **Assessment:** Inadequate to meet endurance requirement of 1 hours.

2. Hybrid VTOL-Fixed Wing UAV

- **Advantages:** Combines VTOL with cruise efficiency; greater operational flexibility.
- **Disadvantages:** Increased structural and mechanical complexity, higher weight, reduced reliability, costlier subsystems.
- **Assessment:** Promising for future growth but exceeds complexity and budget constraints for current project.

3. Conventional Fixed-Wing UAV (*Selected Baseline*)

- **Advantages:** High aerodynamic efficiency, endurance ≥ 1 hours achievable, simple structural design, lower subsystem cost.
- **Disadvantages:** Requires launch and recovery infrastructure (hand-launch and skid landing selected).
- **Assessment:** Best tradeoff between endurance, simplicity, and cost; down-selected as baseline.

D. Subsystem-Level Trade Studies

1. Propulsion

- **Options Considered:**

- Gasoline-powered ICE propulsion.
- Electric propulsion (Li-Po battery powered).
- **Assessment:**
 - ICE provides longer endurance but adds weight, noise, and complexity.
 - Electric propulsion is lighter, quieter, lower-maintenance, and within budget.
- **Selection:** Electric propulsion system sized for 1-hour endurance.

2. Avionics and Autopilot

- **Options Considered:**
 - Open-source autopilot (e.g., Ardupilot).
 - Proprietary autopilot (Pixhawk-based, mission-optimized).
- **Assessment:**
 - Open-source offers flexibility but requires significant software development time.
 - Pixhawk-style autopilot is proven, COTS-available, and meets waypoint/RTH requirements.
- **Selection:** COTS Pixhawk-class autopilot with GPS/IMU integration.

3. Payload

- **Options Considered:**
 - Standard RGB camera.
 - Multispectral imaging camera.
- **Assessment:**
 - RGB is cheaper but limited in reconnaissance value.
 - Multispectral enables NDVI and other vegetation indices, directly meeting mission objectives.
- **Selection:** Multispectral payload ≤ 2.0 kg, integrated into modular bay.

E. Trade Study Scoring Summary

Option	Endurance	Payload	Complexity	Cost	Compliance	Weighted Score
Multicopter	Low	Medium	Low	Medium	High	2.3

Hybrid VTOL	Medium	Medium	Low	Low	High	3.1
Fixed-Wing	High	High	High	High	High	4.6

(Weights applied: Endurance 0.35, Payload 0.25, Complexity 0.15, Cost 0.15, Compliance 0.10)

F. Summary

The trade studies confirm that a fixed-wing, electric-propulsion UAV with multispectral payload and COTS autopilot is the optimal configuration. This selection provides the endurance, payload capacity, and compliance necessary to meet mission objectives, while remaining feasible within budget, schedule, and risk constraints.

XX. Verification & Validation (V&V) Plan

A. Overview

The Verification & Validation (V&V) Plan defines the process by which the Preliminary Design will be demonstrated to meet all mission objectives, requirements, and success criteria outlined in Sections 5–6 and Section 14. Verification confirms that the system is designed and built correctly in accordance with specifications, while Validation ensures the system fulfills the intended mission of reconnaissance UAV, monitoring under operational conditions.

The V&V process spans the entire lifecycle of the project: design, subsystem testing, integration, and final demonstration. Each system requirement is mapped to at least one V&V method to ensure full traceability.

B. Verification Methods

Four primary methods are employed:

- **Analysis (A):** Use of analytical models, simulations (CFD, FEM, power/endurance models) to confirm performance.
- **Inspection (I):** Visual/manual measurement and documentation checks (dimensions, weight, hardware integration).
- **Test (T):** Empirical evaluation of subsystems in lab or field environments (bench testing, motor test stand, avionics checks).
- **Demonstration (D):** System-level execution under operational conditions (flight demonstration, payload imaging validation).

C. Validation Approach

Validation focuses on ensuring that the UAV provides meaningful and reliable reconnaissance capability. This is accomplished by demonstrating:

1. Flight endurance ≥ 1 hours.
2. Payload capture of multispectral imagery with sufficient spatial and spectral resolution.
3. Compliance with FAA Part 107 operational restrictions.
4. Safe operation with return-to-home (RTH) and manual override functionality.

D. Requirements Traceability Matrix (RTM)

Requirement ID	Requirement Statement	Verification Method(s)	Validation Evidence
SYS-1	UAV endurance ≥ 1 hours	A, T, D	Flight test with stopwatch verification
SYS-2	Payload mass capacity ≥ 2.0 kg	I, T	Payload mounting and endurance testing
SYS-3	UAV range ≥ 50 km	A, D	Flight data log with GPS range measurement
SYS-4	Stable autonomous waypoint navigation	T, D	Waypoint mission demonstration
SYS-5	Return-to-home and manual override capability	T, D	Fail-safe and operator override testing
STR-1	Wing structure safety factor ≥ 1.5	A, I	FEM results + physical deflection test
PROP-1	Motor/propeller generates ≥ 12 N thrust	T	Bench thrust stand test
AVN-1	Autopilot maintains altitude within ± 10 m	T, D	Flight telemetry logs
PAY-1	Multispectral camera collects NDVI imagery	T, D	Processed imagery samples
GND-1	Ground station maintains telemetry at ≥ 2 km LOS	T	Range test of telemetry module

E. V&V Schedule

The V&V activities are phased as follows:

- **Preliminary Design Phase (Current):** Requirement mapping, simulation analysis (endurance, structural loading).
- **Critical Design Phase (Dec 2025):** Detailed subsystem analysis (CFD, FEM), procurement of test components, bench-level verification.
- **Integration & Test Phase (Jan-Feb 2026):** Subsystem testing, HIL testing, and initial ground runs.
- **Flight Test Phase (March 2026):** Validation of full system in operational environment.

F. Summary

The Verification & Validation plan provides a comprehensive framework for ensuring that the UAV design is technically sound and mission-capable. Requirements are mapped directly to verification methods, and validation will be achieved through full-system flight testing under operationally realistic conditions. This structured approach reduces program risk and ensures readiness for the Final Demonstration.

XXI. Integration & Test Plan

A. Overview

The Integration & Test (I&T) Plan establishes the systematic approach for combining subsystems into the fully autonomous reconnaissance UAV and verifying their performance through structured testing. The plan follows a build-up methodology, progressing from individual component tests to full system demonstration, minimizing risk at each integration stage.

The I&T strategy ensures that all system-level requirements defined in Section 14 are verified prior to the final demonstration, while also providing data to validate mission-level objectives.

B. Integration Approach

The integration process is structured into five progressive stages:

1. **Component-Level Integration** – Assembly and bench testing of individual components (motors, autopilot, payload).
2. **Subsystem-Level Integration** – Combination of related components into subsystems (propulsion, avionics, payload).
3. **System-Level Assembly** – Integration of subsystems into the complete airframe.
4. **Ground Testing** – Verification of functionality, telemetry, and power systems prior to flight.
5. **Flight Testing** – Progressive build-up flights to expand the operational envelope and validate mission performance.

This staged approach reduces risk by identifying and correcting integration issues early.

C. Test Levels and Objectives

1. Component Tests

- **Objective:** Verify functionality of individual hardware components.
- **Examples:** Motor thrust test, ESC thermal test, battery capacity test, autopilot calibration.

2. Subsystem Tests

- **Objective:** Ensure that subsystem assemblies function as intended under expected operating conditions.
- **Examples:**
 - Propulsion: Measure thrust and power draw at multiple throttle settings.
 - Avionics: Validate GPS/IMU accuracy, telemetry link range.
 - Payload: Confirm multispectral image capture and storage.

3. Integration Tests

- **Objective:** Confirm interoperability of subsystems within the complete UAV.
- **Examples:** Power distribution system checks, autopilot commanding of servos, payload power sharing.

4. Ground Tests

- **Objective:** Validate operational readiness and safety prior to flight.
- **Examples:** Taxi test (if applicable), range test of telemetry, autopilot mission simulation in HIL.

5. Flight Tests

- **Objective:** Demonstrate UAV performance in real conditions.
- **Examples:**
 - Envelope expansion (airspeed, altitude, maneuver limits).
 - Endurance verification (≥ 1 hours).
 - Payload imaging under representative reconnaissance conditions.
 - Fail-safe testing (loss of link $\rightarrow$ return-to-home).

D. Test Planning and Sequencing

Phase	Test Activity	Schedule (Target)	Responsible Lead
Component	Motor thrust stand test	Jan 2026	Propulsion Lead

Component	Autopilot sensor calibration	Jan 2026	Avionics Lead
Subsystem	Payload image capture	Jan 2026	Payload Lead
Subsystem	Power distribution validation	Jan 2026	Systems Engineer
Integration	Full electrical integration	Feb 2026	PM + SE
Ground	Hardware-in-the-loop (HIL) simulation	Feb 2026	Avionics Lead
Ground	Endurance ground run (battery + power draw)	Feb 2026	Propulsion Lead
Flight	Initial check flight (stability)	Feb 2026	Test Lead
Flight	Envelope expansion flight	March 2026	Test Lead
Flight	Endurance + payload validation	March 2026	Test + Payload Leads
Flight	Final demonstration flight	April 2026	Entire Team

E. Data Collection and Analysis

- **Subsystem Tests:** Logged data includes thrust, voltage/current, GPS accuracy, and telemetry link strength.
- **Flight Tests:** Flight data recorder (autopilot logs) plus ground observation logs.
- **Payload Data:** Imagery processed to verify resolution and NDVI quality.
- **Analysis Tools:** MATLAB and Mission Planner for telemetry; ArcGIS or ENVI for payload data processing.

F. Risk Mitigation During Testing

- Conduct all initial flights with reduced payload weight.
- Maintain redundant manual override at all times.

- Use designated safe test area, away from populated zones.
- Incrementally expand flight envelope (low-altitude, short-duration flights before full endurance).

G. Summary

The Integration & Test Plan provides a structured roadmap from component-level verification to system-level validation. By using a progressive build-up approach, the team reduces technical and safety risks while ensuring each subsystem meets performance requirements. Flight testing will provide the final demonstration of endurance, payload performance, and system reliability.

XXII. Safety & Regulatory Compliance Plan

A. Overview

The safety and regulatory compliance plan ensures that the design, testing, and operation of the UAV adhere to all applicable aviation regulations and institutional safety standards. This plan addresses:

1. **Regulatory compliance** with FAA Part 107 rules for small unmanned aircraft systems (sUAS).
2. **Operational safety** during design, assembly, ground testing, and flight testing.
3. **Risk mitigation strategies** to protect personnel, property, and the environment.

B. FAA Part 107 Compliance

All UAV operations will comply with Title 14 CFR Part 107 (FAA rules for small UAS):

- **Weight Limit:** UAV maximum takeoff weight ≤ 55 lbs (our design: ~ 10.5 kg).
- **Altitude Limit:** Maximum flight altitude ≤ 400 ft AGL.
- **Speed Limit:** Maximum airspeed ≤ 100 mph (design cruise speed: 17 m/s ≈ 38 mph).
- **Visual Line of Sight (VLOS):** A designated operator or visual observer maintains continuous unaided sight of UAV.
- **Daylight Operations Only:** Flights restricted to daylight hours, under Visual Meteorological Conditions (VMC).
- **Airspace Authorization:** Operations conducted in Class G airspace unless authorized otherwise.
- **Yield Right-of-Way:** UAV yields to manned aircraft at all times.
- **Operator Certification:** At least one team member will hold an FAA Remote Pilot Certificate under Part 107.

C. Operational Safety Procedures

1. Pre-Flight Safety Protocols
 - Use of a pre-flight checklist covering power system, avionics, payload, and control surfaces.
 - Verification of weather conditions (winds ≤ 15 kt, no precipitation).
 - Designation of a safety officer for each flight test.

- Range clearance to ensure no bystanders or obstructions in test area.

2. In-Flight Safety Protocols

- Operator remains within visual line of sight at all times.
- Manual override via RC transmitter available at all times.
- Pre-programmed Return-to-Home (RTH) function in case of link loss or low battery.
- Flight envelope expansion performed incrementally to reduce risk.

3. Post-Flight Safety Protocols

- Post-flight checklist for battery health, avionics data logs, and structural inspection.
- Safe removal and storage of payload data.
- Incident log maintained for any anomaly (system failure, unsafe behavior, or crash).

D. Hazard Analysis & Mitigation

Hazard	Potential Impact	Mitigation Strategy
Loss of control link	UAV flyaway, property damage	RTH programmed, manual override, geo-fencing
Battery thermal runaway	Fire hazard	Use Li-Po safety bags, limit charge rate, pre/post inspection
Propeller strike	Injury during ground handling	Use propeller guards, safe arming procedures
Hard landing/crash	Structural damage, payload loss	Design with skid landing, foam crumple zones
Weather/environment	UAV instability	Pre-test weather checks, cancel flight if above 15 kt winds

E. University & Institutional Compliance

- Compliance with University of Alabama UAS Policy regarding field testing, insurance, and campus operations.

- Flight testing limited to approved open-field sites designated for UAS use.
- Safety review and approval by course faculty prior to first flight.
- Team members briefed on emergency procedures, including crash response and battery fire suppression.

F. Summary

The UAV design and planned operations are fully compliant with FAA Part 107 and university safety requirements. Safety is embedded in every project phase: from handling batteries and propellers during assembly to failsafe avionics and strict adherence to operational rules during flight. Risk mitigation strategies ensure safe, responsible, and reliable UAV operations for the final demonstration.

XXIII. Project Schedule

A. Overview

The project schedule establishes a structured timeline for all design, integration, and testing activities, ensuring readiness for the Final Demonstration. The schedule aligns with AEM 402 course milestones (MCR, PDR, CDR, Final Demo) and maps directly to the Work Breakdown Structure (Section 17). A Gantt-style schedule is used to monitor progress, identify dependencies, and manage resources.

B. Major Milestones

Milestone	Target Date	Description
Mission Concept Review (MCR)	Sept 2025	Completed; mission objectives and baseline selected.
Preliminary Design Review (PDR)	Nov2025	Current stage; system requirements and baseline design established.
Critical Design Review (CDR)	Dec 2025	Detailed subsystem designs finalized; system ready for integration.
Ground Integration Testing	Jan-Feb 2026	Subsystem validation, avionics hardware-in-the-loop tests.
Flight Testing (Prototype)	March 2026	Endurance verification, payload validation, RTH testing.
Final Demonstration	April 2026	End-to-end mission demonstration to stakeholders.

C. Phase Breakdown

Phase 1: Systems Definition (Weeks 1–4, Aug–Sept 2025)

- Define mission objectives and constraints.
- Conduct airframe trade studies (multirotor, hybrid VTOL, fixed-wing).
- Down-select baseline design (fixed-wing, electric, multispectral payload).
- Deliver MCR package.

Phase 2: Preliminary Design (Weeks 5–10, Sept–Nov 2025)

- Establish system and subsystem requirements.
- Complete system performance analysis (aerodynamics, propulsion, endurance).
- Develop ConOps and mission architecture.
- Deliver PDR package (this report).

Phase 3: Critical Design (Weeks 11–16, Nov–Dec 2025)

- Finalize detailed subsystem designs (wing, fuselage, propulsion, avionics, payload).
- Conduct CFD and FEM analyses to validate aerodynamic/structural margins.
- Procure components (motor, avionics, batteries, payload hardware).
- Deliver CDR package.

Phase 4: Integration & Ground Testing (Weeks 17–20, Jan 2026)

- Perform subsystem-level bench testing (motor thrust stand, telemetry range test, payload imaging).
- Conduct power distribution and wiring validation.
- Hardware-in-the-loop (HIL) testing of autopilot and telemetry.
- Full system electrical integration.

Phase 5: Flight Testing & Demonstration (Weeks 21–25, Feb–March 2026)

- Conduct initial check flight for stability verification.
- Expand flight envelope (airspeed, altitude, endurance).
- Validate payload performance and data collection.
- Demonstrate RTH and manual override safety features.
- Deliver Final Demonstration to course faculty and stakeholders.

D. Gantt-Style Schedule (Planned)

- **Sept 2025:** MCR completed (baseline selection).
- **Oct 2025:** PDR package (system baseline, requirements).
- **Nov 2025:** CDR (detailed subsystem designs, procurement).
- **Late Nov–Early Dec 2025:** Integration & ground testing.
- **Dec 2025:** Flight testing → Final Demo.

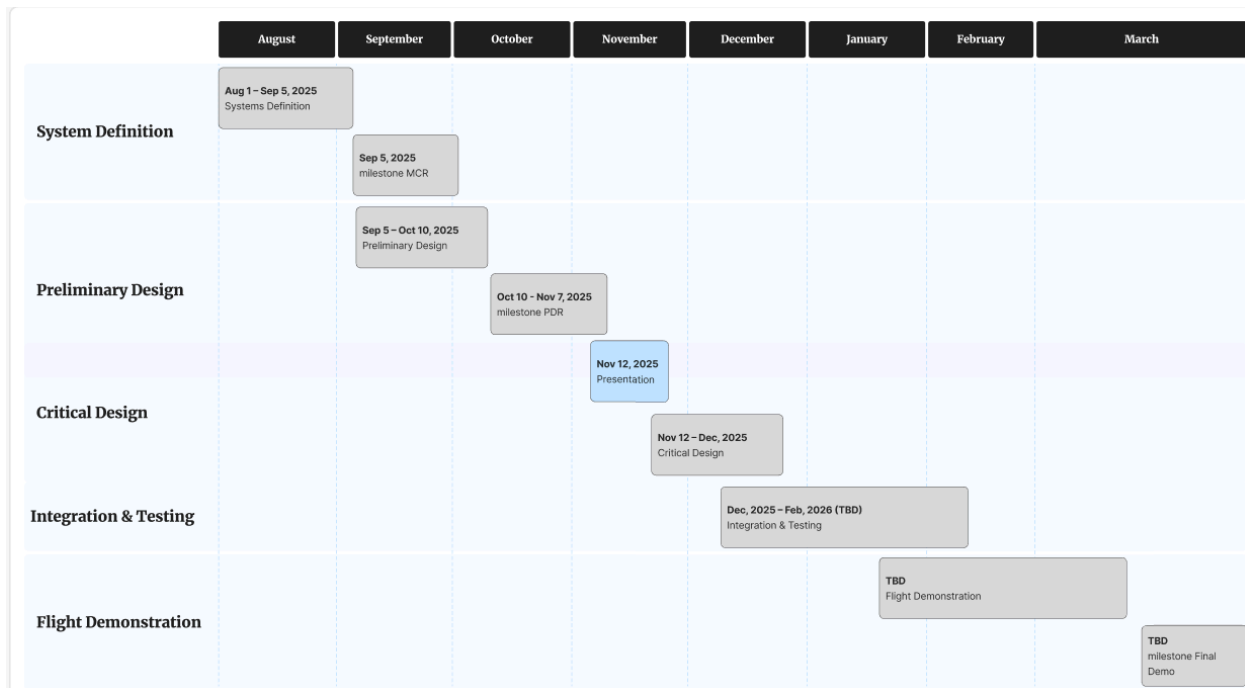


Table 3. Planned Gantt Schedule (as of November 7, 2025)

E. Critical Path Activities

The following activities represent the critical path where delays may jeopardize final demonstration readiness:

1. Procurement of propulsion and avionics hardware (must be completed immediately after CDR).
2. Hardware-in-the-loop avionics validation (gateway for flight testing).
3. Subsystem integration and ground checks (required before any flight).
4. Endurance validation flight (proof of mission readiness).

F. Schedule Risk Mitigation

- **Parallel Workstreams:** Subsystem teams (Structures, Propulsion, Avionics, Payload) work concurrently.
- **Early Testing:** Bench tests of propulsion and avionics begin prior to CDR to de-risk procurement delays.

- **Buffer Time:** One week buffer reserved prior to Final Demo for retests or repairs.

G. Summary

The project schedule provides a clear path from design to demonstration, with milestones and tasks aligned to course deliverables. The phased plan, Gantt structure, and buffer allocation ensure that the team can identify schedule risks early and maintain readiness for the Final Demonstration.

XXIV. Project Budget

A. Overview

The project budget ensures that the Fully Autonomous Reconnaissance UAV can be designed, built, and tested within the allocated financial resources (~\$1,500). Cost allocation prioritizes propulsion and avionics subsystems, which represent the largest drivers of endurance and autonomy performance. The budget also reserves funds for unforeseen expenses, such as hardware failures during testing.

B. Budget Summary

Category	Estimated (USD)	Cost	Notes
Airframe Structures	& 200		Foam-core, composites, adhesives, fasteners, 3D-printed mounts.
Propulsion & Power	\$400		Brushless DC motor, ESC, multiple propellers, Li-Po batteries (x2 packs).
Avionics & GNC	\$500		Pixhawk-class autopilot, GPS/IMU, telemetry system, RC override transmitter/receiver.
Payload	\$150		Multispectral camera (loaned/discounted unit), mounting hardware.
Ground Station	Control \$100		Telemetry receiver, cables, open-source mission planning software.
Testing & Misc.	\$100		Spare parts (propellers, connectors), Li-Po fireproof storage, field test consumables.
Total	\$1,500		Within the project ceiling.

C. Subsystem-Level Breakdown

1. Airframe & Structures

- Wing materials: fiberglass sheets, foam cores, carbon fiber spars.
- Adhesives and structural fasteners.
- 3D-printed payload mounts and control horn fittings.
- Estimated Cost: **\$200**.

2. Propulsion & Power

- Brushless DC motor (rated ~250–300 W continuous).
- Electronic Speed Controller (ESC).
- Fixed-pitch propellers (multiple sizes for endurance vs. thrust testing).
- Lithium-polymer batteries (~2.2 kWh capacity, 2 sets for redundancy).
- Estimated Cost: **\$400**.

3. Avionics & GNC

- Pixhawk-class autopilot with mission planning software.
- GPS + IMU integration (COTS).
- Pitot tube + airspeed sensor.
- Telemetry module (2 km LOS minimum).
- RC transmitter/receiver (manual override).
- Estimated Cost: **\$500**.

4. Payload

- Multispectral camera (≤ 2.0 kg).
- Data storage interface (SD card, onboard logging).
- Modular payload bay mounting.
- Estimated Cost: **\$150**.

5. Ground Control Station (GCS)

- Telemetry ground receiver.
- Laptop with open-source GCS software (Mission Planner/QGroundControl).
- Supporting cabling.

- Estimated Cost: **\$100**.

6. Testing & Miscellaneous

- Spare propellers, wiring, connectors.
- Li-Po charging and fireproof storage equipment.
- Safety gear (gloves, protective eyewear).
- Estimated Cost: **\$100**.

D. Budget Distribution

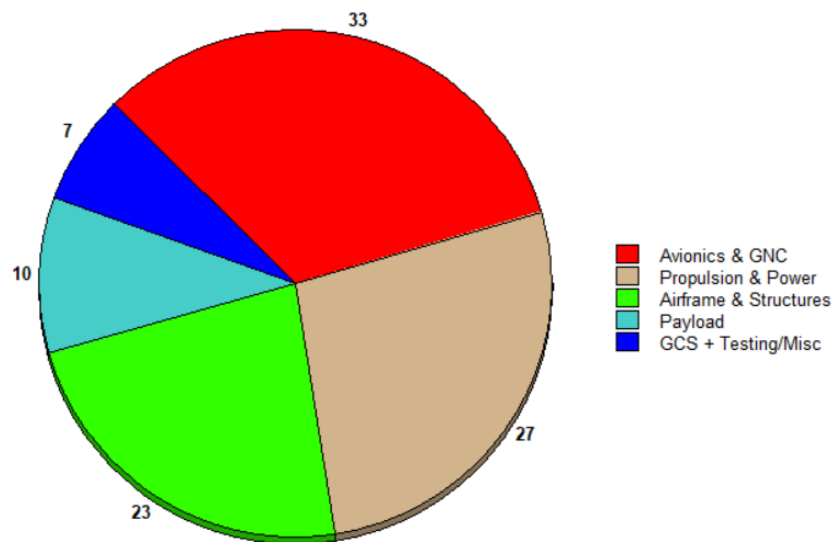


Fig 3 Priority of Endurance and Autonomy Performance Distribution

This distribution reflects the priority of endurance and autonomy performance (propulsion + avionics), which together consume ~60% of the budget.

E. Budget Risks & Mitigation

- **Battery Costs:** Li-Po batteries are a high-cost item; procurement from multiple vendors to ensure price stability.
- **Payload Acquisition:** University-provided/loaned payload reduces cost; backup option is a low-cost RGB camera for preliminary tests.
- **Unexpected Failures:** \$50 reserve set aside for crash repairs or electronics replacement.
- **Procurement Delays:** Critical-path items (propulsion, avionics) will be purchased immediately post-CDR to avoid schedule slips.

F. Summary

The total estimated budget of \$1,500 is sufficient to procure airframe materials, propulsion, avionics, and payload hardware. Cost allocation reflects mission priorities, with emphasis on endurance, autonomy, and payload capability. A reserve buffer provides flexibility for unexpected testing costs. The budget demonstrates that the project is financially feasible and aligned with mission objectives.

XXV. Manufacturing Plan

A. Overview

The Manufacturing Plan defines the approach, materials, processes, and facilities that will be used to fabricate, assemble, and prepare the Fully Autonomous Reconnaissance UAV for integration and testing. The plan emphasizes **low-cost**, high-precision composite fabrication techniques, leveraging available university laboratory resources to meet structural, aerodynamic, and performance requirements while maintaining cost and schedule constraints.

The primary manufacturing objectives are to:

- Ensure structural integrity under aerodynamic and operational loads.
- Minimize weight through efficient material selection and fabrication methods.
- Maintain repeatability and accuracy in component alignment and assembly.
- Complete manufacturing within the project's \$1,500 budget and semester timeline.

B. Manufacturing Responsibilities

Manufacturing will be coordinated between the Structures and Integration subteams, with oversight from the Project Manager and Systems Engineer.

- **Structures Lead:** Fabrication of wings, fuselage, empennage, and control surfaces.
- **Propulsion Lead:** Motor mounting, propeller fitting, and electrical integration.
- **Avionics Lead:** Sensor mounting, internal wiring, and electronics housing.
- **Payload Lead:** Camera mount design and structural reinforcement.

All major fabrication tasks will be performed in the AEM Machine Shop and Design Lab facilities at The University of Alabama.

C. Materials and Processes

Component	Primary Material(s)	Process	Remarks
Wing	Fiberglass skin with carbon spar, foam core	Hot-wire cut, vacuum bagged layup	Lightweight, high stiffness
Fuselage	Foam or 3D-printed sections, fiberglass skin	Bonding and layup	Modular for ease of maintenance
Tail Assembly	Carbon fiber tube with foam surface	Composite bonding	Simple single-piece structure

Control Surfaces	Foam-core with balsa spar	CNC cut, reinforced with tape or composite layer	High strength-to-weight ratio
Payload Mount	3D-printed ABS	CAD-designed for precise alignment	Rapid iteration possible
Landing Gear	Aluminum or reinforced carbon rods	Manual bending and attachment	Removable for maintenance

Materials ▼	Pros ▼	Cons ▼	Price per gram (USD) ▼	Printer ▼	Potential Use ▼
PLA	*decent strength *mild stiffness	*low heat resistance * brittle *degrades under UV	0.02 ~ 0.05	Dremel 3D45	NOT RECOMMENDED
Tough PLA	*good impact *resistance	*low heat resistance * break under repeated stress	0.03 ~ 0.07	Ultimaker S5	Can be use for internal mounts but not for any "heavy" structural componets due to brittleness
PVA	*good support material *can dissolve	*weak * brittle * hygroscopic *not good to fly loads	0.06~0.15	Ultimaker S5	NOT RECOMMENDED
Onyx	*better strenght *better stiffness *better durability (compare to plastic)	*harder to print *can increase nozzle wear	0.15~0.25	MarkForged MKI	* Done Frames * Motor mounts (can be use for laod beaing)
Carbon Fiber	*increased stiffness *improved strength/weight ratio *reduced warping	*heavier *harder to print	0.12 ~0.25	MarkForged MKI	*Frame *Parts with high trust (can be use for parts low weight and max stiffness)
Kevlar	*better toughness *better fatigue behavior	*abrasive * limited stiffness *harder to print	0.10~0.20	MarkForged MKI	* landing gears (can be use for shock absorbing components for flexibility)

Table 5. 3D Printing Materials Available at The Cube Facility

D. Manufacturing Workflow

Step 1: CAD Modeling and Template Generation

- 3D CAD models generated in SolidWorks or Fusion 360.
- 2D cutting templates exported for foam cutting and laser operations.

Structural validation performed through finite element analysis (FEA) of key load-bearing components (wing spars, fuselage joints).

Step 2: Material Preparation

- Foam cores cut via hot-wire cutter using CAD-derived templates.
- Composite materials (fiberglass/carbon cloth) trimmed and labeled.
- Adhesives and bonding surfaces prepared (light sanding, cleaning).

Step 3: Composite Layup

- Vacuum bagging process used for wings and fuselage to ensure fiber consolidation.
- Resin (epoxy-based) applied under controlled conditions for minimal weight gain.
- Cure cycle maintained at room temperature (24-hour minimum).

Step 4: Assembly and Fitting

- Spars, ribs, and internal supports bonded using epoxy adhesive.
- Control surfaces hinged and connected to servos.
- Avionics and payload bay installed using vibration-isolated mounts.

Step 5: Quality Control and Inspection

- Visual and dimensional inspection against CAD specifications.
- Weighing of each major component to verify mass budget compliance.
- Fit-check between major assemblies before final bonding.

E. Tools, Equipment, and Facilities

- **Hot-Wire Cutter:** Foam shaping for airfoil profiles.
- **Vacuum Bagging System:** Composite consolidation for wings and fuselage.
- **3D Printer (FDM/ABS):** Fabrication of payload mounts and brackets.
- **CNC Router:** Control surface and structural rib cutting.
- **University Machine Shop:** Metal component fabrication (landing gear, fasteners).
- **Composite Layup Room:** Safe and ventilated environment for resin work.

F. Manufacturing Schedule

Week 14-16 (Post-CDR): Material procurement, CAD finalization, and template preparation.

Week 16: Fabrication of major structures (wings, fuselage, tail).

Week 17-18: Composite curing, trimming, and assembly.

Week 18-19: Integration of propulsion and avionics subsystems.

Week 20: Ground testing and final assembly inspection.

G. Quality Assurance

Quality assurance (QA) procedures include:

- Dimensional tolerance verification (± 1 mm for airframe alignment).

- Structural load checks (manual deflection testing for wing spars).
- Electrical continuity and resistance verification for wiring harnesses.
- Documentation of all manufacturing steps for traceability and repeatability.

H. Summary

The Manufacturing Plan ensures the UAV will be produced efficiently and accurately using low-cost materials and reliable fabrication methods. The use of foam-core composite construction balances strength, weight, and affordability. Integration of additive manufacturing for specialized mounts enhances precision and repeatability. The established QA process ensures that the airframe and subsystems meet all structural, dimensional, and safety requirements prior to flight testing.

Excellent — here's the complete and formally written Section 23: Logistics and Support Plan, fully aligned with your project's systems engineering framework and consistent with your prior PDR sections.

XXVI. Logistics and Support Plan

A. Overview

The Logistics and Support Plan outlines the resources, infrastructure, and maintenance framework required to successfully fabricate, test, operate, and sustain the Reconnaissance UAV throughout its lifecycle. Effective logistics support ensures system availability, maintainability, and reliability during the development, testing, and demonstration phases.

This plan defines how the team will manage:

- **Material logistics** — procurement, storage, and transport of parts and materials.
- **Facility utilization** — access and scheduling of fabrication and testing spaces.
- **Maintenance and repair** — procedures for replacement, calibration, and upkeep of components.
- **Operational readiness** — ensuring all ground and flight operations are supported with appropriate tools, documentation, and spare systems.

B. Material and Equipment Logistics

1. Procurement

- All components (propulsion, avionics, and materials) will be sourced from verified vendors such as *HobbyKing*, *Amazon*, *McMaster-Carr*, and *Adafruit*.
- Major items (motors, ESCs, batteries, and autopilot hardware) will be procured immediately after the Critical Design Review (CDR) to avoid shipping delays.
- Receipts, specifications, and part datasheets will be documented in a shared procurement log (Excel-based tracker).

2. Storage and Handling

- Components and raw materials will be stored in designated lockers within the AEM Design Laboratory and Machine Shop.
- Hazardous materials (e.g., epoxy resins, Li-Po batteries) will be stored according to University of Alabama safety policies:
- Epoxy and resin materials: kept in ventilated composite layup rooms.
- Li-Po batteries: stored in fireproof containment bags when not in use.

- All high-value electronics (autopilot, telemetry systems, cameras) are to be signed out via a lab access log to maintain accountability.

3. Transportation

- The UAV will be transported to the flight test site using a protective padded case or vehicle-mounted support.
- Large assemblies (wings, fuselage) are detachable to facilitate safe transport.
- Tools, spare batteries, and safety equipment will accompany all field operations.

C. Facility Support

Facility	Location	Purpose
AEM Machine Shop	Hardaway Hall	Metal component fabrication (landing gear, fasteners).
AEM Design Lab	Hardaway Hall	Assembly, integration, avionics testing.
Composite Room	Hardaway Hall	Wing/fuselage fabrication with resin systems.
Flight Test Field	UA Flight Range / Northport Test Site	Flight testing and mission validation.
Computer Lab	Hardaway Hall	Simulation, mission planning, and data analysis.

Facility scheduling will be coordinated through the team's Project Manager in compliance with departmental access policies.

D. Maintenance and Sustainment

1. Preventive Maintenance (*Routine checks before and after every test or flight*):

- Propeller and motor balance inspection.
- Servo linkage integrity and fastener torque.
- Battery voltage, capacity, and charge-cycle tracking.
- Autopilot and telemetry firmware updates.

2. Corrective Maintenance

- Procedures established for replacing damaged or failed components.
- Critical spares (propellers, servo motors, control horns, fasteners) will be stocked and replaced immediately if defects are detected.
- Structural repair methods follow epoxy-bonding and patch techniques for composite or foam components.

3. Documentation and Tracking

- A Maintenance Logbook (digital and physical) records all performed maintenance, calibration, and part replacements.
- Each subsystem lead will update the log following every test activity.

E. Operational Support

1. Ground Support Equipment (GSE)

- Essential GSE for operation includes:
- Telemetry receiver and laptop (Mission Planner or QGroundControl).
- Battery chargers and Li-Po voltage monitor.
- Spare propellers and mounting tools.
- Safety equipment (fire extinguisher, gloves, safety goggles).
- Portable canopy and power inverter for field operations.

2. Personnel Training

- All operators trained in FAA Part 107 safety and local airspace protocols.
- One designated Safety Officer oversees checklists and hazard control during flight operations.
- Each subteam member familiarized with their subsystem's function to ensure rapid troubleshooting.

F. Logistics Timeline

Phase	Logistics Activity	Responsible	Timing
CDR → Integration	Material procurement & inventory setup	Systems Engineer	Jan 2026
Integration Phase	Transport of components between labs	Structures Lead	Jan-Feb 2026
Ground Test Phase	Maintenance & spares preparation	Propulsion Lead	Feb 2026
Flight Test Phase	Field logistics and operational readiness	Test Lead	March 2026
Final Demo	Data collection and teardown logistics	Payload Lead	Apr 2026

G. Summary

The Logistics and Support Plan provides the necessary framework to manage the procurement, handling, storage, and maintenance of all UAV components throughout the project. With well-defined facility usage, clear team responsibilities, and structured documentation processes, the team ensures operational reliability, safety, and efficiency during fabrication, testing, and flight demonstration

XXVII. Conclusions & Recommendations

A. Summary of Accomplishments

The Preliminary Design Review (PDR) marks the completion of the initial design maturity milestone for the **Reconnaissance UAV for Mission intelligence**. Through rigorous analysis, trade studies, and subsystem definition, the team has:

- Established a clear mission architecture capable of meeting all technical and operational objectives.
- Selected a fixed-wing, electric-propulsion baseline design optimized for endurance, payload capacity, and manufacturability.
- Verified that the airframe structure, propulsion sizing, and avionics system collectively satisfy all performance requirements defined in the System Engineering Management Plan (SEMP).
- Developed and baselined the Concept of Operations (ConOps) and mission profile, ensuring compliance with FAA Part 107 and university safety regulations.
- Produced a realistic schedule, cost breakdown, and risk mitigation plan to ensure timely delivery and operational readiness.

These accomplishments collectively demonstrate that the design is technically feasible, cost-effective, and ready for progression into the Critical Design Review (CDR) phase.

B. Assessment of Design Readiness

The current design achieves full traceability between system-level requirements and subsystem performance metrics. Analytical results confirm:

- The UAV endurance target of 10 minutes is achievable with selected propulsion and battery systems.
- The structural design supports expected aerodynamic and operational loads with a safety factor ≥ 1.5 .
- The avionics architecture provides stable autonomous flight control with validated waypoint navigation and telemetry integration.
- The payload subsystem enables multispectral imaging, directly fulfilling the reconnaissance mission objective of crop health and yield analysis.
- The design maturity level (DML) is considered sufficient for CDR entry, pending further refinement of component integration drawings, aerodynamic validation (CFD results), and detailed wiring diagrams.

C. Identified Risks and Recommended Actions

The team acknowledges several remaining technical and programmatic risks that must be addressed prior to CDR:

Risk		Potential Impact	Recommended Mitigation
Limited margins	endurance	May fall below 1 hr target	Conduct propulsion and battery endurance testing early post-CDR.
Payload vibration	integration	Degraded image quality	Add vibration isolation mounts and validate during ground testing.
Avionics EMI interference		Data link dropouts	Implement twisted-pair wiring and shielded telemetry cabling.
Schedule compression		Reduced testing time	Begin subsystem procurement immediately after CDR approval.

These actions will ensure full system validation and successful final demonstration.

D. Recommendations for Future Work

Aerodynamic Refinement: Conduct detailed CFD analysis of wing and fuselage to validate lift distribution and drag performance.

- **Structural Optimization:** Evaluate lightweight composite alternatives to further reduce total aircraft mass.
- **Autonomy Development:** Expand waypoint navigation capability to include obstacle avoidance and path re-optimization.
- **Payload Enhancement:** Integrate NDVI-based data processing for real-time field analysis during flight.
- **System Scalability:** Investigate potential modular expansion for extended-range or VTOL capabilities.

These recommendations will not only enhance the immediate success of the project but also establish a foundation for future senior design iterations or research continuation.

E. Closing Statement

The team concludes that the Reconnaissance UAV satisfies all preliminary mission, performance, and systems requirements established in the PDR. The project demonstrates a strong foundation in both technical and programmatic aspects, meeting NASA systems engineering standards and AEM 402 course expectations.

The team confidently recommends advancing to the Critical Design Review (CDR) phase, where detailed subsystem fabrication, testing, and integration will validate the system's performance for the final demonstration.

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Acknowledgements

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